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(54) **SYSTEMS AND METHODS FOR
CUSTOMIZING MECHANICAL STRENGTH
IN STIMULI-RESPONSIVE
BIOABSORBABLE MATERIALS**

(58) **Field of Classification Search**
CPC C08G 63/08; A61L 31/146; A61L 31/148;
A61L 27/18; A61L 27/56; A61L 31/06;
(Continued)

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(21) Appl. No.: **18/484,929**

Pattnaik, Ananya, et al. "Designing of Gradient Scaffolds and Their
Applications in Tissue Regeneration." *Biomaterials*, vol. 296, May
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Primary Examiner — Nathaniel C Chukwurah

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Related U.S. Application Data

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(57) **ABSTRACT**

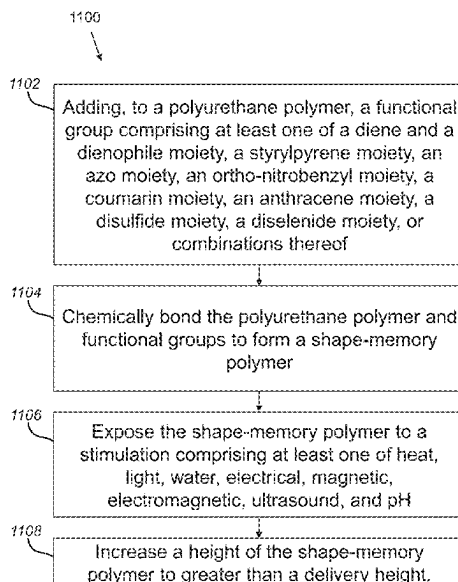
(51) **Int. Cl.**
A61B 17/064 (2006.01)
A61B 17/068 (2006.01)

(Continued)

The disclosed technology includes a bioabsorbable material
configured to be delivered to tissue. The material includes a
shape-memory polymer compressible in a delivery configu-
ration and configured to swell within a predetermined period
of time. The shape-memory polymer includes one or more
functional groups for reversible bonding between adjacent
functional groups to transition between an approximately
linear polymer and an approximately non-linear polymer
upon exposure to a stimulation.

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25 Claims, 9 Drawing Sheets



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- (58) **Field of Classification Search**
- CPC *A61L 27/58*; *C08J 9/26*; *C08J 9/36*; *C08J 2207/10*; *C08J 2201/0482*; *C08J 2367/04*; *C08J 2205/02*; *C08J 2201/048*; *C08J 2300/16*; *C08J 2205/05*; *C08L 67/04*
- See application file for complete search history.
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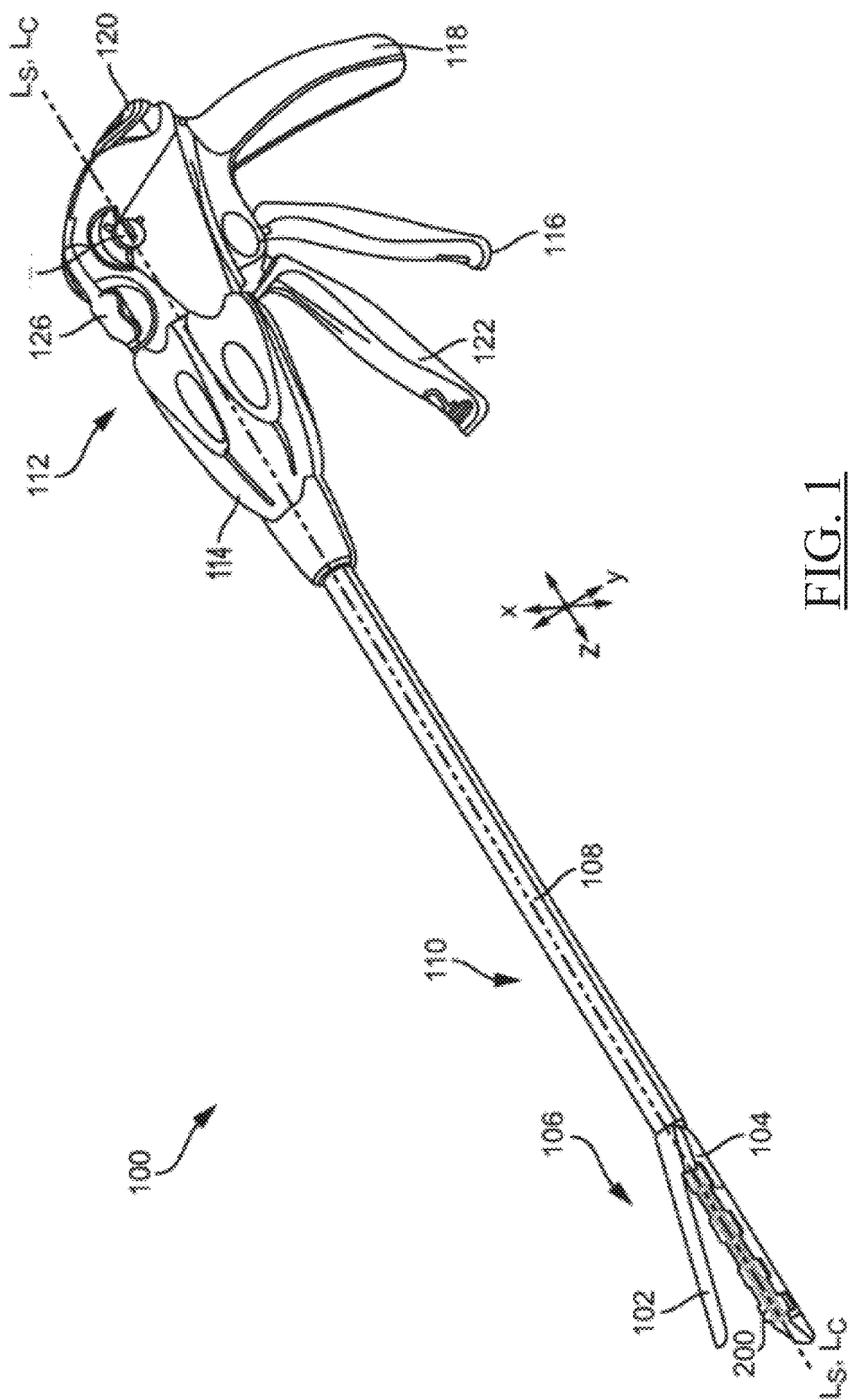


FIG. 1

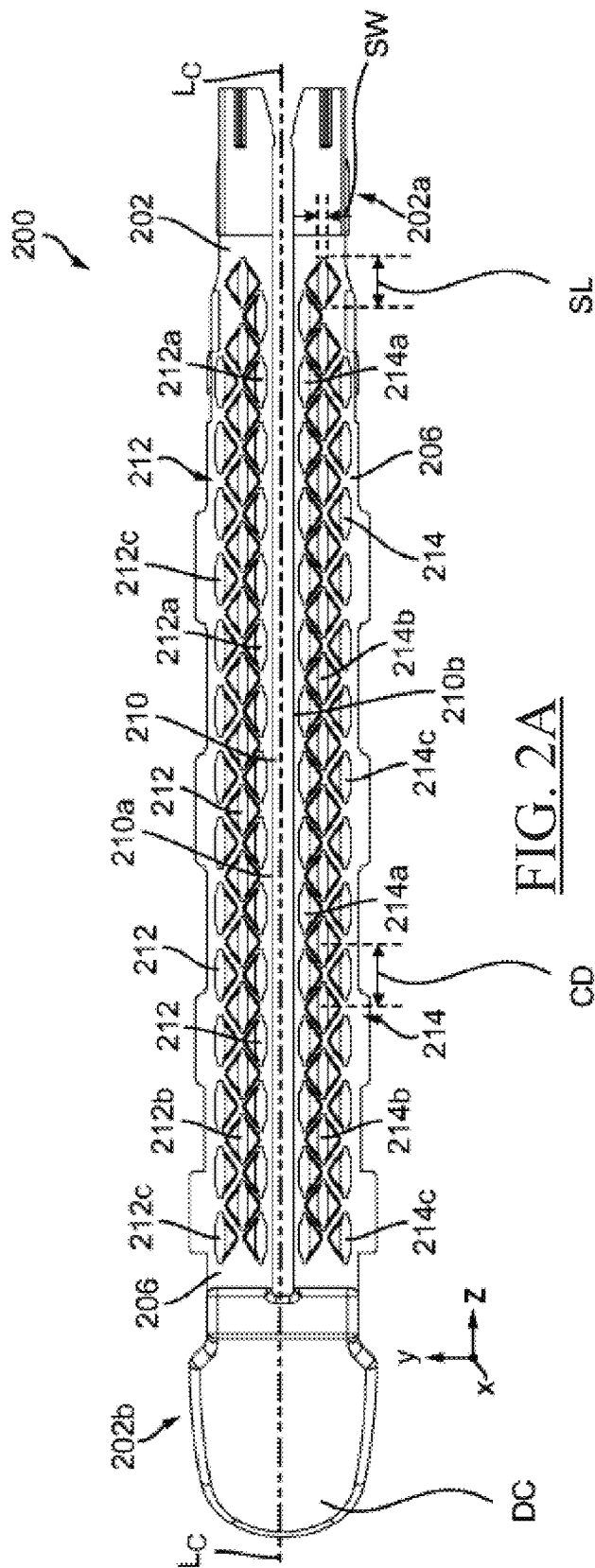


FIG. 2A

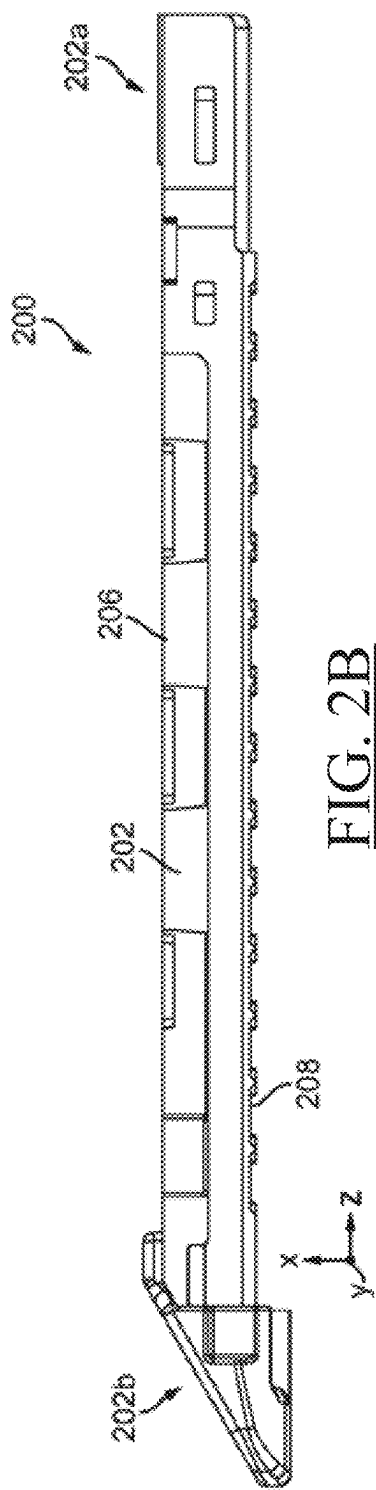


FIG. 2B

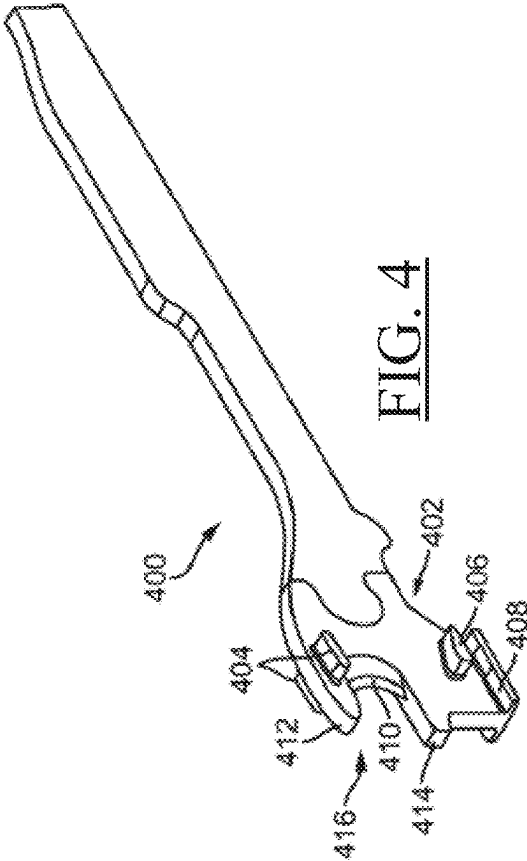


FIG. 4

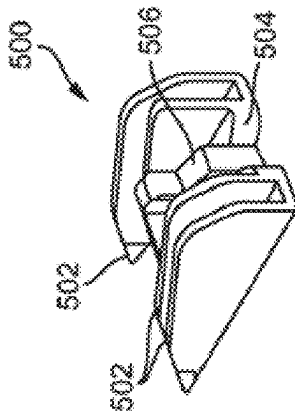


FIG. 5

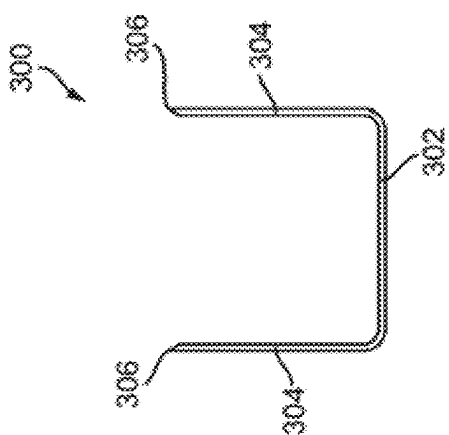


FIG. 3

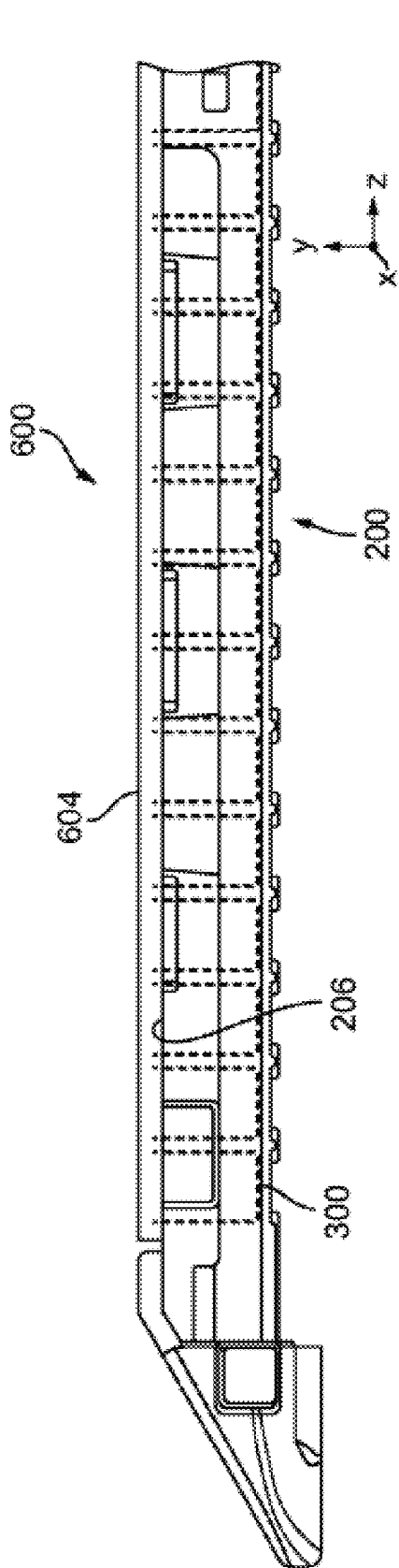


FIG. 6A

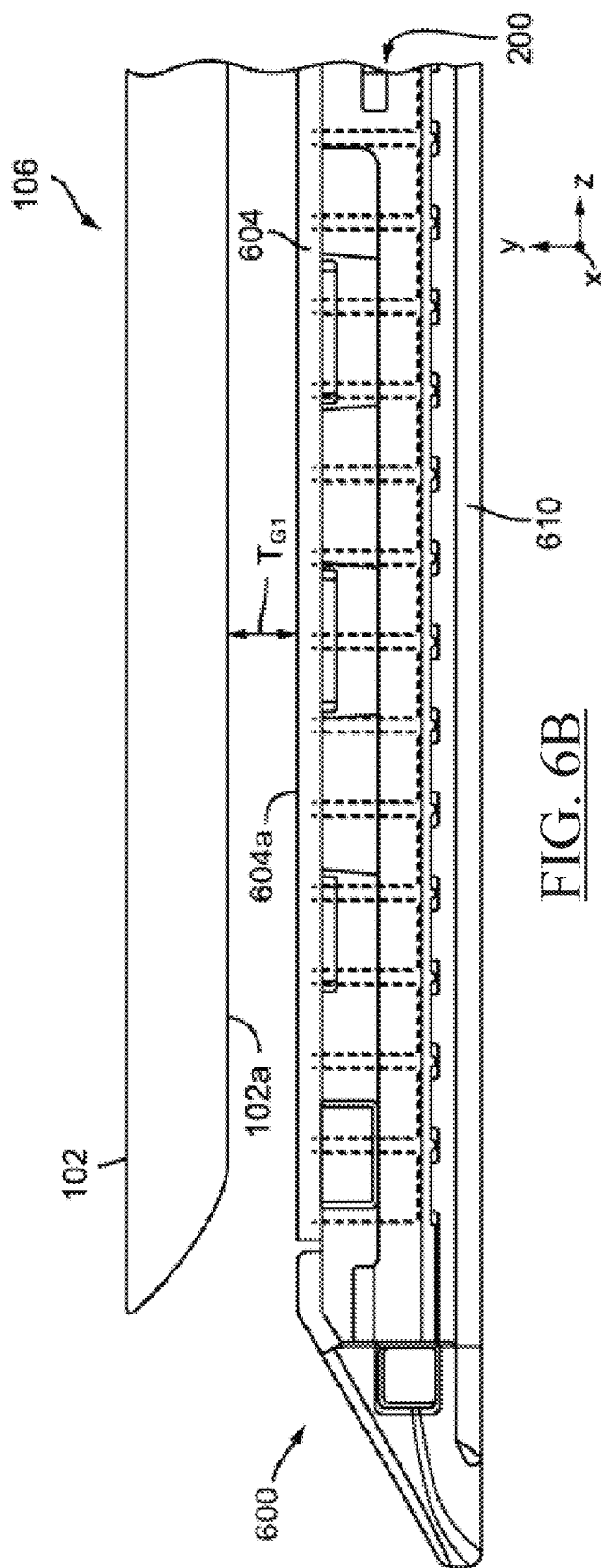
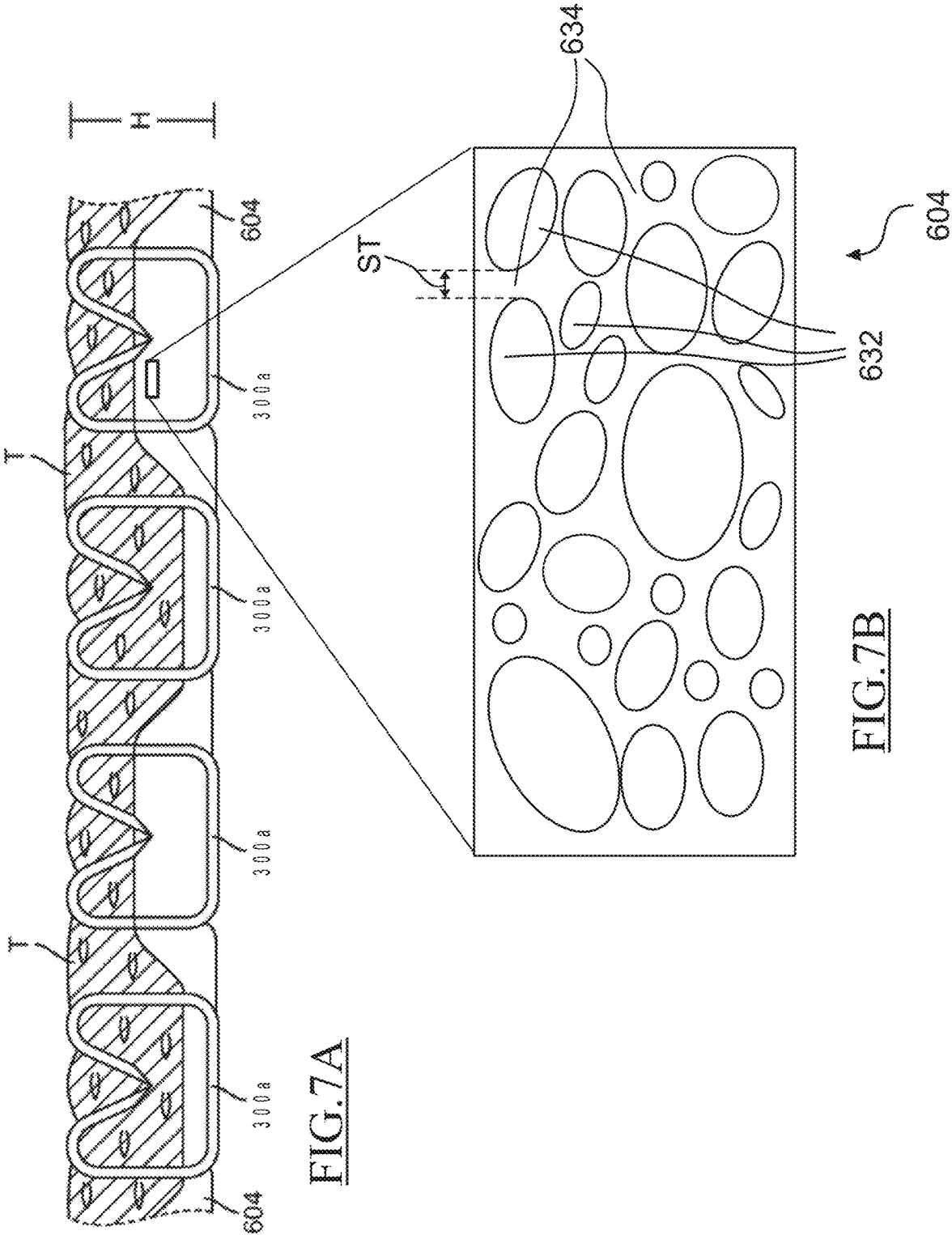


FIG. 6B



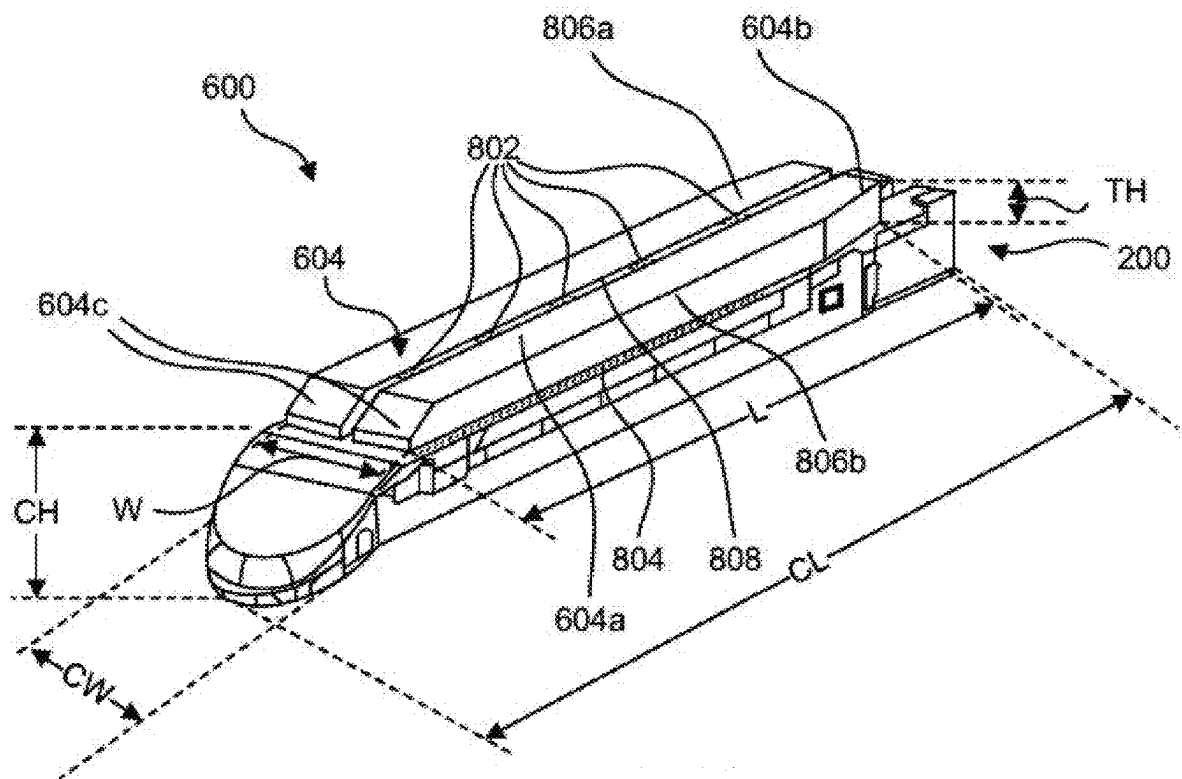
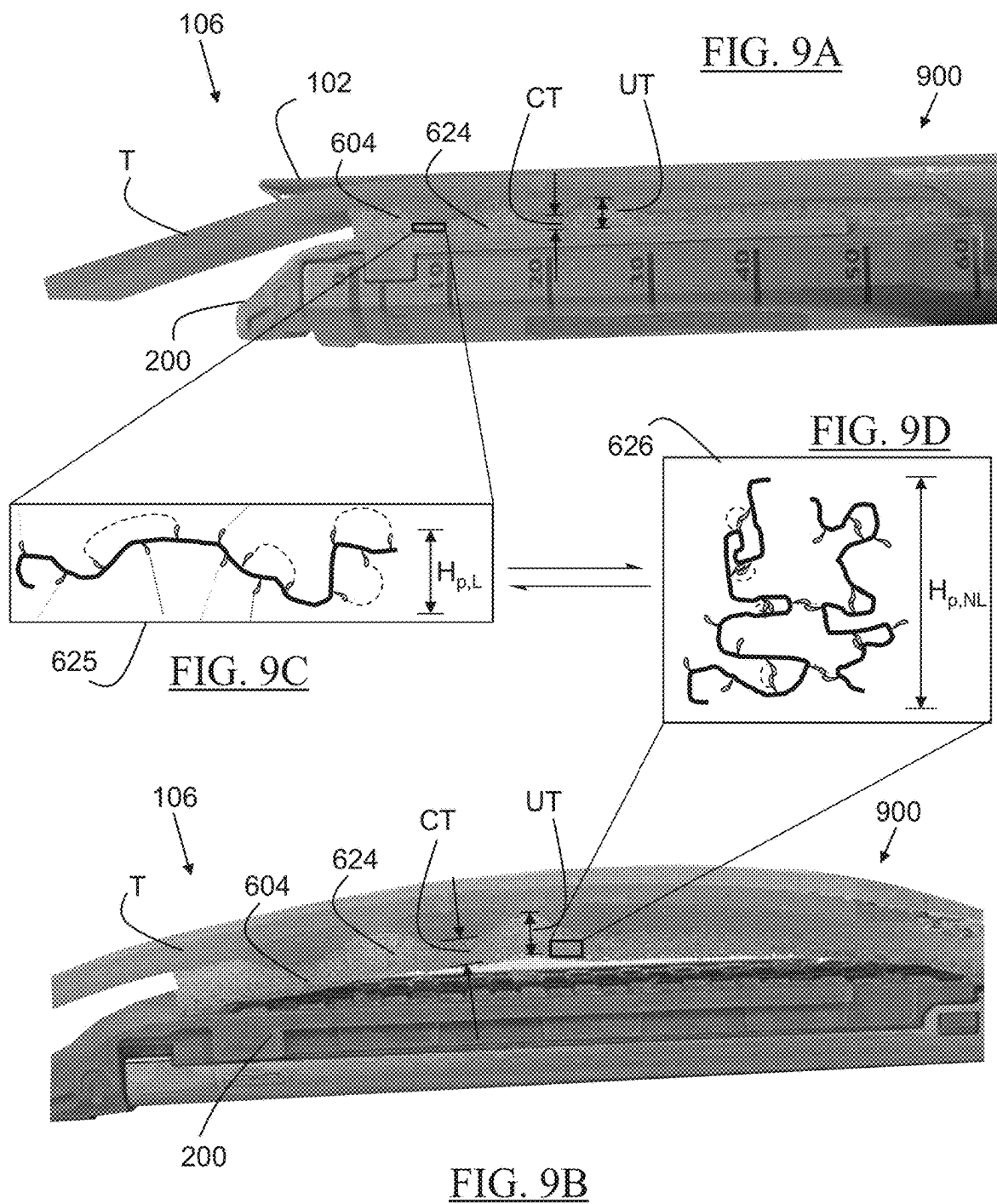


FIG. 8



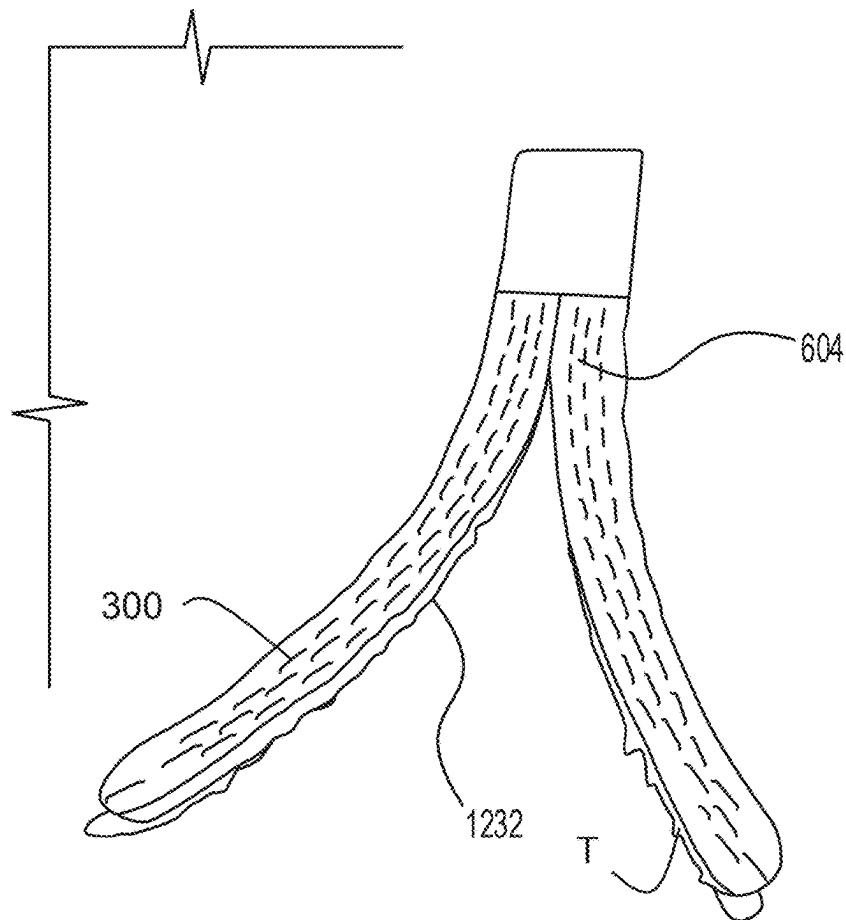


FIG. 10A

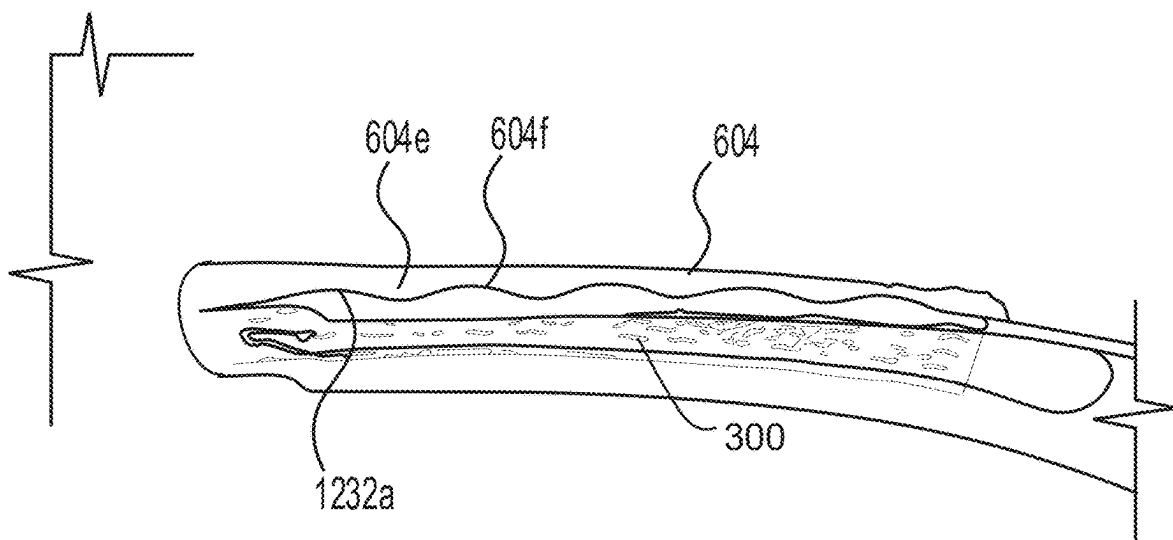
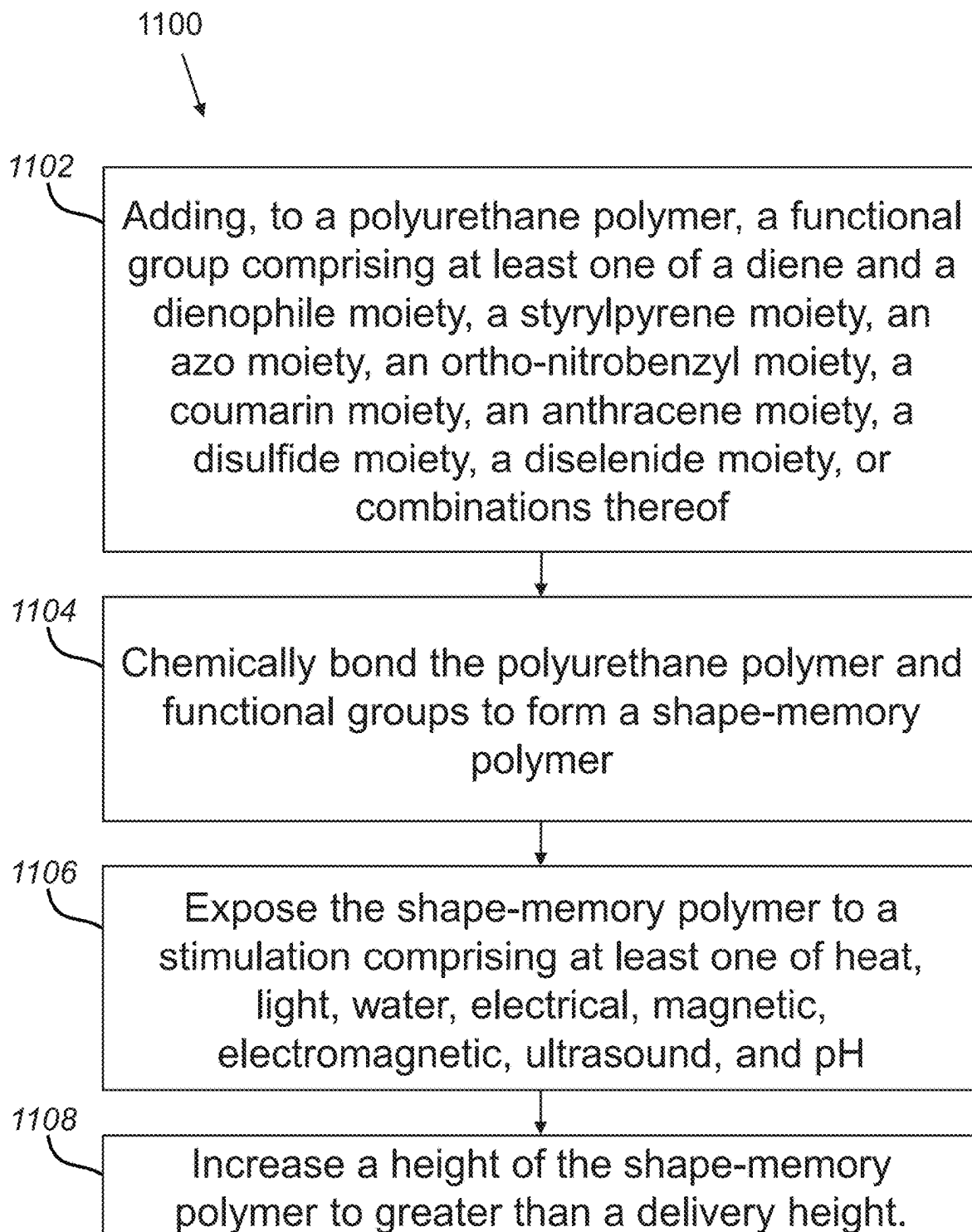


FIG. 10B

FIG. 11

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SYSTEMS AND METHODS FOR CUSTOMIZING MECHANICAL STRENGTH IN STIMULI-RESPONSIVE BIOABSORBABLE MATERIALS

CROSS-REFERENCE TO RELATED APPLICATIONS

This application claims priority under 35 U.S.C. § 119 to U.S. Provisional Patent Application No. 63/522,660, filed Jun. 22, 2023, the entire contents of which are fully incorporated herein by reference.

FIELD

The present invention relates generally to systems and methods for customizing mechanical properties via stimuli-responsive functional groups in bioabsorbable materials.

BACKGROUND

Surgical staplers are used in surgical procedures to close openings in tissue, blood vessels, ducts, shunts, or other objects or body parts involved in the particular procedure. The openings can be naturally occurring, such as passageways in blood vessels or an internal organ like the stomach, or they can be formed by the surgeon during a surgical procedure, such as by puncturing tissue or blood vessels to form a bypass or an anastomosis, or by cutting tissue during a stapling procedure.

Most staplers have a handle (some of which are directly user operable, others of which are operable by a user via a robotic interface) with an elongate shaft extending from the handle and having a pair of movable opposed jaws formed on an end thereof for holding and forming staples therebetween. The staples are typically contained in a staple cartridge, which can house multiple rows of staples and is often disposed in one of the two jaws for ejection of the staples to the surgical site. In use, the jaws are positioned so that the object to be stapled is disposed between the jaws, and staples are ejected and formed when the jaws are closed, and the device is actuated. Some staplers include a knife configured to travel between rows of staples in the staple cartridge to longitudinally cut and/or open the stapled tissue between the stapled rows.

SUMMARY

There is provided, in accordance with an example of the present invention, a bioabsorbable material configured to be delivered to tissue. The material includes a shape-memory polymer compressible in a delivery configuration and configured to swell within a predetermined period of time. The shape-memory polymer includes one or more functional groups for reversible bonding between adjacent functional groups to transition between an approximately linear polymer and an approximately non-linear polymer upon exposure to a stimulation.

There is provided, in accordance with an exemplary embodiment of the present invention, a shape-memory polymer compressible in a delivery configuration and configured to swell within a predetermined period of time upon exposure to a stimulation. The shape-memory polymer can include a polyurethane backbone and one or more functional groups for reversible bonding between adjacent functional groups to transition between an approximately linear polymer to an approximately non-linear polymer.

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There is provided, in accordance with an example of the present invention, a method to form a bioabsorbable material. The method can include the steps of adding, to a polyurethane polymer, a functional group and chemically bonding the polyurethane polymer and functional groups to form a shape-memory polymer. The functional group can include at least one of a diene and a dienophile moiety, a styrylpyrene moiety, an azo moiety, an ortho-nitrobenzyl moiety, a coumarin moiety, an anthracene moiety, a disulfide moiety, a diselenide moiety, or combinations thereof.

BRIEF DESCRIPTION OF THE DRAWINGS

This invention will be more fully understood from the following detailed description taken in conjunction with the accompanying drawings, in which:

FIG. 1 is a perspective view of one exemplary embodiment of a conventional surgical stapling and severing instrument.

FIG. 2A is a top view of a staple cartridge for use with the surgical stapling and severing instrument of FIG. 1;

FIG. 2B is a side view of the staple cartridge of FIG. 2A;

FIG. 3 is a side view of a staple in an unfired (pre-deployed) configuration that can be disposed within the staple cartridge of the surgical cartridge assembly of FIG. 2A;

FIG. 4 is a perspective view of a knife and firing bar ("E-beam") of the surgical stapling and severing instrument of FIG. 1;

FIG. 5 is a perspective view of a wedge sled of a staple cartridge of the surgical stapling and severing instrument of FIG. 1;

FIG. 6A is a longitudinal cross-sectional view of an exemplary surgical cartridge assembly having a compressible non-fibrous adjunct attached to a top or deck surface of a staple cartridge;

FIG. 6B is a longitudinal cross-sectional view of a surgical end effector having an anvil pivotably coupled to an elongate channel and the surgical cartridge assembly of FIG. 6A disposed within and coupled to the elongate channel, showing the anvil in a closed position without any tissue between the anvil and the adjunct;

FIG. 7A is a partial-schematic illustrating the adjunct of FIGS. 6A-6B in a tissue deployed condition;

FIG. 7B is a diagram showing an enlarged portion of an exemplary adjunct with a porous structure;

FIG. 8 is a perspective view of an exemplary cartridge assembly;

FIG. 9A is a side view of an exemplary end effector having an adjunct of bioabsorbable material in a delivery configuration;

FIG. 9B is a side view of an exemplary end effector having an adjunct of bioabsorbable material after firing and release from cartridge;

FIGS. 9C and 9D illustrate example reversible transformations from linear polymers to complex architectures;

FIG. 10A is a top perspective view of an exemplary adjunct after use;

FIG. 10B is a side view of an exemplary adjunct after use; and

FIG. 11 is a flow chart showing an exemplary method of forming a surgical adjunct with spatial control of mechanical properties in bioabsorbable material.

DETAILED DESCRIPTION

The following detailed description should be read with reference to the drawings, in which like elements in different

drawings are identically numbered. The drawings, which are not necessarily to scale, depict selected embodiments and are not intended to limit the scope of the invention. The detailed description illustrates by way of example, not by way of limitation, the principles of the invention. This description will clearly enable one skilled in the art to make and use the invention, and describes several embodiments, adaptations, variations, alternatives and uses of the invention, including what is presently believed to be the best mode of carrying out the invention.

Ranges may be expressed herein as from "about" or "approximately" or "substantially" one particular value and/or to "about" or "approximately" or "substantially" another particular value. When such a range is expressed, other exemplary embodiments include from the one particular value and/or to the other particular value. Further, the term "about" means within an acceptable error range for the particular value as determined by one of ordinary skill in the art, which will depend in part on how the value is measured or determined, i.e., the limitations of the measurement system. For example, "about" can mean within an acceptable standard deviation, per the practice in the art. Alternatively, "about" can mean a range of up to $\pm 20\%$, preferably up to $\pm 10\%$, more preferably up to $\pm 5\%$, and more preferably still up to $\pm 1\%$ of a given value. Alternatively, particularly with respect to biological systems or processes, the term can mean within an order of magnitude, preferably within 2-fold, of a value. Where particular values are described in the application and claims, unless otherwise stated, the term "about" is implicit and in this context means within an acceptable error range for the particular value.

In addition, throughout this disclosure, various aspects of the disclosure can be presented in a range format. It should be understood that the description in range format is merely for convenience and brevity and should not be construed as an inflexible limitation on the scope of the disclosure. Accordingly, the description of a range should be considered to have specifically disclosed all the possible subranges as well as individual numerical values within that range. For example, description of a range such as from 1 to 6 should be considered to have specifically disclosed subranges such as from 1 to 3, from 1 to 4, from 1 to 5, from 2 to 4, from 2 to 6, from 3 to 6 etc., as well as individual numbers within that range, for example, 1, 2, 2.7, 3, 4, 5, 5.3, and 6. This applies regardless of the breadth of the range.

Similarly, as used herein, "substantially free" of something, or "substantially pure", and like characterizations, can include both being "at least substantially free" of something, or "at least substantially pure", and being "completely free" of something, or "completely pure."

By "comprising" or "containing" or "including" is meant that at least the named compound, element, particle, or method step is present in the composition or article or method, but does not exclude the presence of other compounds, materials, particles, method steps, even if the other such compounds, material, particles, method steps have the same function as what is named.

The term "aliphatic" or "aliphatic group," as used herein, means a straight-chain (i.e., unbranched) or branched, substituted or unsubstituted hydrocarbon chain that is completely saturated or that contains one or more units of unsaturation, or a monocyclic hydrocarbon or bicyclic hydrocarbon that is completely saturated or that contains one or more units of unsaturation, but which is not aromatic (also referred to herein as "carbocycle" or "cycloaliphatic"), that has a single point of attachment to the rest of the molecule. Unless otherwise specified, aliphatic groups contain 1-6

aliphatic carbon atoms. In some embodiments, aliphatic groups contain 1-5 aliphatic carbon atoms. In other embodiments, aliphatic groups contain 1-4 aliphatic carbon atoms. In still other embodiments, aliphatic groups contain 1-3 aliphatic carbon atoms, and in yet other embodiments, aliphatic groups contain 1-2 aliphatic carbon atoms. Suitable aliphatic groups include, but are not limited to, linear or branched, substituted or unsubstituted alkyl, alkenyl, alkynyl groups and hybrids thereof such as cycloalkyl, (cycloalkyl)alkyl, (cycloalkenyl)alkyl or (cycloalkyl)alkenyl.

The term "alkyl," unless otherwise indicated, as used herein, refers to a monovalent aliphatic hydrocarbon radical having a straight chain, branched chain, monocyclic moiety, or polycyclic moiety or combinations thereof, wherein the radical is optionally substituted at one or more carbons of the straight chain, branched chain, monocyclic moiety, or polycyclic moiety or combinations thereof with one or more substituents at each carbon, wherein the one or more substituents are independently C_1 - C_{10} alkyl. In some embodiments, "cycloalkyl" (or "carbocycle") refers to a monocyclic C_3 - C_6 hydrocarbon that is completely saturated or that contains one or more units of unsaturation, but which is not aromatic, that has a single point of attachment to the rest of the molecule. A carbocycle can be, under certain circumstances, a bridged bicyclic or a fused ring such as, e.g., an ortho-fused carbocycle, a spirofused carbocycle, etc. Examples of "alkyl" groups include, but are not limited to, methyl, ethyl, propyl, isopropyl, butyl, iso-butyl, sec-butyl, tert-butyl, pentyl, hexyl, heptyl, cyclopropyl, cyclobutyl, cyclopentyl, cyclohexyl, cycloheptyl, norbornyl, and the like.

The term "alkylene" refers to a bivalent alkyl group. An "alkylene chain" is a polymethylene group, i.e., $-(CH_2)_n-$, wherein n is a positive integer, preferably from 1 to 6, from 1 to 4, from 1 to 3, from 1 to 2, or from 2 to 3. A substituted alkylene chain is a polymethylene group in which one or more methylene hydrogen atoms are replaced with a substituent. Suitable substituents include those described below for a substituted aliphatic group.

The term "alkenylene" refers to a bivalent alkenyl group. A substituted alkenylene chain is a polymethylene group containing at least one double bond in which one or more hydrogen atoms are replaced with a substituent. Suitable substituents include those described below for a substituted aliphatic group.

The term "alkoxy" means an alkyl radical attached through an oxygen linking atom, represented by $-O$ -alkyl. For example, " (C_1-C_4) alkoxy" includes methoxy, ethoxy, propoxy, and butoxy.

The term "aryl," used alone or as part of a larger moiety, refers to monocyclic or bicyclic ring systems having a total of five to fourteen ring members, wherein at least one ring in the system is aromatic and wherein each ring in the system contains 3 to 7 ring members. The term "aryl" may be used interchangeably with the term "aryl ring." In certain embodiments of compounds described herein, "aryl" refers to an aromatic ring system which includes, but is not limited to, phenyl, naphthyl, anthracyl and the like, which may bear one or more substituents. It will be appreciated that an "aryl" group can comprise carbon and heteroatom ring members.

As described herein, compounds described herein may contain "optionally substituted" moieties. In general, the term "substituted," whether preceded by the term "optionally" or not, means that one or more hydrogens of the designated moiety are replaced with a suitable substituent. Unless otherwise indicated, an "optionally substituted" group may have a suitable substituent at each substitutable

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position of the group, and when more than one position in any given structure may be substituted with more than one substituent selected from a specified group, the substituent may be either the same or different at every position. Combinations of substituents envisioned by this disclosure are preferably those that result in the formation of stable or chemically feasible compounds. The term “stable,” as used herein, refers to compounds that are not substantially altered when subjected to conditions to allow for their production, detection, and, in certain embodiments, their recovery, purification, and use for one or more of the purposes disclosed herein.

The term “polyurethane,” as used herein, refers to a polymeric reaction product of an isocyanate and a polyol, and is not limited to those polymers which include only urethane or polyurethane linkages. It is well understood by those of ordinary skill in the art of preparing polyurethanes that the polyurethane polymers may also include linkages such as allophanate, carbodiimide, and other linkages described herein in addition to urethane linkages.

The expressions “reaction system,” “reactive formulation,” “reaction product,” and “reactive mixture” are interchangeably used herein, and all refer to a combination of reactive compounds used to make the bioabsorbable material according to the disclosure.

The term “room temperature” refers to temperatures of about 20° C., this means referring to temperatures in the range 18° C. to 25° C. Such temperatures will include 18° C., 19° C., 20° C., 21° C., 22° C., 23° C., 24° C. and 25° C.

Unless otherwise expressed, the “weight percentage” (indicated as % wt. or wt. %) of a component in a composition refers to the weight of the component over the total weight of the composition in which it is present and is expressed as percentage.

“Glass transition temperature” and “ T_g ” as referred to herein refers to the temperature at which a reversible transition from a hard glass condition into a rubber-elastic condition occurs.

Surgical stapling assemblies and methods for manufacturing and using the same are provided. In general, a surgical stapling assembly can include a staple cartridge having staples disposed therein and an adjunct configured to be releasably retained on the staple cartridge. As discussed herein, the various adjuncts provided can be configured to compensate for variations in tissue properties, such as variations in tissue thickness, and/or to promote tissue ingrowth when the adjuncts are stapled to tissue. As discussed herein, an adjunct can include a bioabsorbable material, such as a foam.

An exemplary stapling assembly can include a variety of features to facilitate application of a surgical staple, as described herein and illustrated in the drawings. However, a person skilled in the art will appreciate that the stapling assembly can include only some of these features and/or it can include a variety of other features known in the art. The stapling assemblies described herein are merely intended to represent certain exemplary examples. Moreover, while the adjuncts are described in connection with surgical staple cartridge assemblies, the adjuncts can be used in connection with staple reloads that are not cartridge based or any type of surgical instrument.

FIG. 1 illustrates an exemplary surgical stapling and severing device **100** suitable for use with an implantable adjunct. The illustrated surgical stapling and severing device **100** includes end effector **106** having an anvil **102** that is pivotably coupled to an elongate channel **104**. As a result, the end effector **106** can move between an open position, as

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shown in FIG. 1, and a closed position in which the anvil **102** is positioned adjacent to the elongate channel **104** to engage tissue therebetween. The end effector **106** can be attached at its proximal end to an elongate shaft **108** forming an implement portion **110**. When the end effector **106** is closed, or at least substantially closed, (e.g., the anvil **102** moves from the open position in FIG. 1 toward the elongate channel) the implement portion **110** can present a sufficiently small cross-section suitable for inserting the end effector **106** through a trocar. While the device **100** is configured to staple and sever tissue, surgical devices configured to staple but not sever tissue are also contemplated herein.

In various instances, the end effector **106** can be manipulated by a handle **112** connected to the elongate shaft **108**. The handle **112** can include user controls such as a rotation knob **114** that rotates the elongate shaft **108** and the end effector **106** about a longitudinal axis (Ls) of the elongate shaft **108** and an articulation control **115** that can articulate the end effector **106** about an articulate axis (TA) that is substantially transverse to the longitudinal axis (Ls) of the elongate shaft **108**. Further controls include a closure trigger **116** which can pivot relative to a pistol grip **118** to close the end effector **106**. A closure release button **120** can be outwardly presented on the handle **112** when the closure trigger **116** is clamped such that the closure release button **120** can be depressed to unclamp the closure trigger **116** and open the end effector **106**, for example. Handle **112** may also take the form of an interface for connection to a surgical robot.

In some examples, a firing trigger **122**, which can pivot relative to the closure trigger **116**, can cause the end effector **106** to simultaneously sever and staple tissue clamped therein. The firing trigger **122** may be powered, require force from a user to engage, or some combination thereof. A manual firing release lever **126** can allow the firing system to be retracted before full firing travel has been completed, if desired, and, in addition, the firing release lever **126** can allow a surgeon, or other clinician, to retract the firing system in the event that the firing system binds and/or fails.

Additional details on the surgical stapling and severing device **100** and other surgical stapling and severing devices suitable for use with the present disclosure are described, for example, in U.S. Pat. No. 9,332,984 and in U.S. Patent Publication No. 2009/0090763, the disclosures of which are incorporated herein by reference in their entireties. Further, the surgical stapling and severing device need not include a handle, but instead can have a housing that is configured to couple to a surgical robot, for example, as described in U.S. Patent Publication No. 2019/0059889, the disclosure of which is incorporated herein by reference in its entirety.

As further shown in FIG. 1, a staple cartridge **200** can be utilized with the instrument **100**. In use, the staple cartridge **200** is placed within and coupled to the elongate channel **104**. While the staple cartridge **200** can have a variety of configurations, in this illustrated example, the staple cartridge **200**, which is shown in more detail in FIGS. 2A-2B, has a proximal end **202a** and a distal end **202b** with a cartridge longitudinal axis (LC) extending therebetween. As a result, when the staple cartridge **200** is inserted into the elongate channel **104** (FIG. 1), the longitudinal axis (LC) is substantially or approximately parallel with the longitudinal axis (LS) of the elongate shaft **108**. Further, the staple cartridge **200** includes a longitudinal slot **210** defined by two opposing walls **210a**, **210b** and configured to receive at least a portion of a firing member of a firing assembly, like firing assembly **400** in FIG. 4, as discussed further below. As

shown, the longitudinal slot **210** extends from the proximal end **202a** toward the distal end **202b** of the staple cartridge **200**. It is also contemplated herein that in other examples, the longitudinal slot **210** can be omitted.

The illustrated staple cartridge **200** includes staple cavities **212**, **214** defined therein, in which each staple cavity **212**, **214** is configured to removably house at least a portion of a staple (not shown). The number, shape, and position of the staple cavities can vary and can depend at least on the size and shape (e.g., mouth-like shape) of the staples to be removably disposed therein. In this illustrated example, the staple cavities are arranged in two sets of three longitudinal rows, in which the first set of staple cavities **212** is positioned on a first side of the longitudinal slot **210** and the second set of staple cavities **214** is positioned on a second side of the longitudinal slot **210**. On each side of the longitudinal slot **210**, and thus for each set of rows, a first longitudinal row of staple cavities **212a**, **214a** extends alongside the longitudinal slot **210**, a second row of staple cavities **212b**, **214b** extends alongside the first row of staple cavities **212a**, **214a**, and a third row of staple cavities **212c**, **214c** extends alongside the second row of staple cavities **212b**, **214b**. Each row may be approximately parallel and the staple cavities that make up the rows may be approximately parallel in orientation with the longitudinal slot **210**. As shown in FIGS. 2A, each staple cavity **212**, **214** may include a maximum length SL of about 0.122 inches to about 0.124 inches and a maximum width SW of about 0.023 inches to about 0.027 inches. In addition, at least the centers of two adjacent cavities **212**, **214** are spaced apart by about 0.158 inches.

The staples releasably stored in the staple cavities **212**, **214** can have a variety of configurations. An exemplary staple **300** that can be releasably stored in each of the staple cavities **212**, **214** is illustrated in FIG. 3 in its unfired (pre-deployed, unformed) configuration. The illustrated staple **300** includes a crown (base) **302** and two legs **304** extending from each end of the crown **302**. In this example, the crown **302** extends in a linear direction and the staple legs **304** have the same unformed height. Further, prior to the staples **300** being deployed, the staple crowns **302** can be supported by staple drivers that are positioned within the staple cartridge **200** and, concurrently, the staple legs **304** can be at least partially contained within the staple cavities **212**, **214**. Further, the staple legs **304** can extend beyond a top surface, like top surface **206**, of the staple cartridge **200** when the staples **300** are in their unfired positions. In certain instances, as shown in FIG. 3, the tips **306** of the staple legs **304** can be pointed and sharp which can incise and penetrate tissue.

In use, staples **300** can be deformed from an unfired position into a fired position such that the staple legs **304** move through the staple cavities **212**, **214**, penetrate tissue positioned between the anvil **102** and the staple cartridge **200**, and contact the anvil **102**. As the staple legs **304** are deformed against the anvil **102**, the legs **304** of each staple **300** can capture a portion of the tissue within each staple **300** and apply a compressive force to the tissue. Further, the legs **304** of each staple **300** can be deformed downwardly toward the crown **302** of the staple **300** to form a staple entrapment area in which the tissue can be captured therein. In various instances, the staple entrapment area can be defined between the inner surfaces of the deformed legs and the inner surface of the crown of the staple. The size of the entrapment area for a staple can depend on several factors such as the length

of the legs, the diameter of the legs, the width of the crown, and/or the extent in which the legs are deformed, for example.

In some examples, all of the staples disposed within the staple cartridge **200** can have the same unfired (pre-deployed, unformed) configuration. In other examples, the staples can include at least two groups of staples each having a different unfired (pre-deployed, unformed) configuration, e.g., varying in height and/or shape, relative to one another, etc.

Referring back to FIGS. 2A-2B, the staple cartridge **200** extends from a top surface or deck surface **206** to a bottom surface **208**, in which the top surface **206** is configured as a tissue-facing surface and the bottom surface **208** is configured as a channel-facing surface. As a result, when the staple cartridge **200** is inserted into the elongate channel **104**, as shown in FIG. 1, the top surface **206** faces the anvil **102** and the bottom surface **208** (obstructed) faces the elongate channel **104**.

With reference to FIGS. 4 and 5, a firing assembly such as, for example, firing assembly **400**, can be utilized with a surgical stapling and severing device, like device **100** in FIG. 1. The firing assembly **400** can be configured to advance a wedge sled **500** having wedges **502** configured to deploy staples from the staple cartridge **200** into tissue captured between an anvil, like anvil **102** in FIG. 1, and a staple cartridge, like staple cartridge **200** in FIG. 1. Furthermore, an E-beam **402** at a distal portion of the firing assembly **400** may fire the staples from the staple cartridge. During firing, the E-beam **402** can also cause the anvil to pivot towards the staple cartridge, and thus move the end effector from the open position towards a closed position. The illustrated E-beam **402** includes a pair of top pins **404**, a pair of middle pins **406**, which may follow a portion **504** of the wedge sled **500**, and a bottom pin or foot **408**. The E-beam **402** can also include a sharp cutting edge **410** configured to sever the captured tissue as the firing assembly **400** is advanced distally, and thus towards the distal end of the staple cartridge. In addition, integrally formed and proximally projecting top guide **412** and middle guide **414** bracketing each vertical end of the cutting edge **410** may further define a tissue staging area **416** assisting in guiding tissue to the sharp cutting edge **410** prior to being severed. The middle guide **414** may also serve to engage and fire the staples within the staple cartridge by abutting a stepped central member **506** of the wedge sled **500** that effects staple formation by the end effector **106**.

In use, the anvil **102** in FIG. 1 can be moved into a closed position by depressing the closure trigger in FIG. 1 to advance the E-beam **402** in FIG. 4. The anvil **102** can position tissue against at least the top surface **206** of the staple cartridge **200** in FIGS. 2A-2B. Once the anvil has been suitably positioned, the staples **300** in FIG. 3 disposed within the staple cartridge can be deployed.

To deploy staples from the staple cartridge, as discussed above, the sled **500** in FIG. 5 can be moved from the proximal end toward a distal end of the cartridge body, and thus, of the staple cartridge. As the firing assembly **400** in FIG. 4 is advanced, the sled can contact and lift staple drivers within the staple cartridge upwardly within the staple cavities **212**, **214**. In at least one example, the sled and the staple drivers can each include one or more ramps, or inclined surfaces, which can co-operate to move the staple drivers upwardly from their unfired positions. As the staple drivers are lifted upwardly within their respective staple cavities, the staples are advanced upwardly such that the staples emerge from their staple cavities and penetrate into

tissue. In various instances, the sled can move several staples upwardly at the same time as part of a firing sequence.

As indicated above, the stapling device can be used in combination with a compressible adjunct. A person skilled in the art will appreciate that, while adjuncts are shown and described below, the adjuncts disclosed herein can be used with other surgical instruments and need not be coupled to a staple cartridge as described. Further, a person skilled in the art will also appreciate that the staple cartridges need not be replaceable.

As discussed above, with some surgical staplers, a surgeon is often required to select the appropriate staples having the appropriate staple height for tissue to be stapled. For example, a surgeon will utilize tall staples for use with thick tissue and short staples for use with thin tissue. In some instances, however, the tissue being stapled does not have a consistent thickness and thus, the staples cannot achieve the desired fired configuration for every section of the stapled tissue (e.g., thick and thin tissue sections). The inconsistent thickness of tissue can lead to undesirable leakage and/or tearing of tissue at the staple site when staples with the same or substantially greater height are used, particularly when the staple site is exposed to intra-pressures at the staple site and/or along the staple line.

Accordingly, various examples of adjuncts are provided that can be configured to compensate for varying thickness of tissue that is captured within fired (deployed) staples to avoid the need to take into account staple height when stapling tissue during surgery. That is, the adjuncts described herein can allow a set of staples with the same or similar heights to be used in stapling tissue of varying thickness (e.g., from thin to thick tissue) while also, in combination with the adjunct, providing adequate tissue compression within and between fired staples. Thus, the adjuncts described herein can maintain suitable compression against thin or thick tissue stapled thereto to thereby minimize leakage and/or tearing of tissue at the staple sites. In addition, exemplary adjuncts described herein may be configured to be absorbed in the body over a period of 100 to 300 days depending on implanted location and tissue health.

Alternatively, or in addition, the adjuncts can be configured to promote tissue ingrowth. In various instances, it is desirable to promote the ingrowth of tissue into an implantable adjunct, to promote the healing of the treated tissue (e.g., stapled and/or incised tissue), and/or to accelerate the patient's recovery. More specifically, the ingrowth of tissue into an implantable adjunct may reduce the incidence, extent, and/or duration of inflammation at the surgical site. Tissue ingrowth into and/or around the implantable adjunct may, for example, manage the spread of infections at the surgical site. The ingrowth of blood vessels, especially white blood cells, for example, into and/or around the implantable adjunct may fight infections in and/or around the implantable adjunct and the adjacent tissue. Tissue ingrowth may also encourage the acceptance of foreign matter (e.g., the implantable adjunct and the staples) by the patient's body and may reduce the likelihood of the patient's body rejecting the foreign matter. Rejection of foreign matter may cause infection and/or inflammation at the surgical site.

In general, the adjuncts provided herein are designed and positioned atop a staple cartridge, like staple cartridge 200. When the staples are fired (deployed) from the cartridge, the staples penetrate through the adjunct and into tissue. As the legs of the staple are deformed against the anvil that is positioned opposite the staple cartridge, the deformed legs capture a portion of the adjunct and a portion of the tissue

within each staple. That is, when the staples are fired into tissue, at least a portion of the adjunct becomes positioned between the tissue and the fired staple. While the adjuncts described herein can be configured to be attached to a staple cartridge, it is also contemplated herein that the adjuncts can be configured to mate with other instrument components, such as an anvil of a surgical stapler. A person of ordinary skill will appreciate that the adjuncts provided herein can be used with replaceable cartridges or staple reloads that are not cartridge based.

In various embodiments, the adjunct or bioabsorbable materials disclosed herein can be comprised of an absorbable polymer. In certain embodiments, an adjunct can be comprised of foam, film, fibrous woven, fibrous non-woven polyurethane, polyether urethane, polyester urethane, polyester urea, polyester, polycarbonate, polyorthoester, polyanhydride, polyesteramide, polyphosphazenes, polyphosphoesters, polysaccharides, and/or polyoxaester. In other embodiments, an adjunct can be a copolymer including, for example, PGA (polyglycolic acid), PGA/PCL (poly(glycolic acid-co-caprolactone)), PLA/PCL (poly(lactic acid-co-poly-caprolactone)), PLLA/PCL, PGA/TMC (poly(glycolic acid-co-trimethylene carbonate)), PDS, PEPBO, and the like. In various embodiments, an adjunct can include an organic material such as, for example, carboxymethyl cellulose, sodium alginate, hyaluronic acid, and/or oxidized regenerated cellulose. In various embodiments, an adjunct has a durometer in the 3-7 Shore A (30-50 Shore OO) ranges with a maximum stiffness of 15 Shore A (65 Shore OO). In certain embodiments, an adjunct can undergo 40% compression under 3 lbf load, 60% compression under 6 lbf load, and/or 80% compression under 20 lbf load, for example. In certain embodiments, one or more gasses, such as air, nitrogen, carbon dioxide, and/or oxygen, for example, can be bubbled through and/or contained within the adjunct.

Methods of Stapling Tissue

FIGS. 6A-6B illustrate an exemplary example of a stapling assembly 600 that includes a staple cartridge 200 and an adjunct 604. For sake of simplicity, the adjunct 604 is generally illustrated in FIGS. 6A-6B, and various configurations of the adjunct are described in more detail below. As shown, the adjunct 604 is positioned against the staple cartridge 200. While partially obstructed in FIGS. 6A-6B, the staple cartridge 200 includes staples 300, that are configured to be deployed into tissue. The staples 300 can have any suitable unformed (pre-deployed) height.

In the illustrated example, the adjunct 604 can be mated to at least a portion of the top surface or deck surface 206 of the staple cartridge 200. In some examples, the top surface 206 of the staple cartridge 200 can include one or more surface features which can be configured to engage the adjunct 604 to avoid undesirable movements of the adjunct 604 relative to the staple cartridge 200 and/or to prevent premature release of the adjunct 604 from the staple cartridge 200. Exemplary surface features are described further below and in U.S. Pat. No. 10,052,104, which is incorporated by reference herein in its entirety.

FIG. 6B shows the stapling assembly 600 placed within and coupled to the elongate channel 610 of surgical end effector 106. The anvil 102 is pivotally coupled to the elongate channel 610 and is thus moveable between open and closed positions relative to the elongate channel 610, and thus the staple cartridge 200. The anvil 102 is shown in a closed position in FIG. 6B and illustrates a tissue gap T_{G1} created between the staple cartridge 602 and the anvil 612. More specifically, the tissue gap T_{G1} is defined by the distance between the tissue-compression surface 102a of the

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anvil **102** (e.g., the tissue-engaging surface between staple forming pockets in the anvil) and the tissue-contacting surface **604a** of the adjunct **604**. In this illustrated example, both the tissue-compression surface **102a** of the anvil **102** and the tissue-contacting surface **604a** of the adjunct **604** is planar, or substantially planar (e.g., planar within manufacturing tolerances). As a result, when the anvil **102** is in a closed position, as shown in FIG. 6B, the tissue gap T_{G1} is generally uniform (e.g., nominally identical within manufacturing tolerances) when no tissue is disposed therein. In other words, the tissue gap T_{G1} is generally constant (e.g., constant within manufacturing tolerances) across the end effector **106** (e.g., in the y-direction). In other examples, the tissue-compression surface of the anvil can include a stepped surface having longitudinal steps between adjacent longitudinal portions, and thus create a stepped profile (e.g., in the y-direction). In such examples, the tissue gap T_{G1} can be varied.

The adjunct **604** is compressible to permit the adjunct to compress to varying heights to thereby compensate for different tissue thickness that are captured within a deployed staple. The adjunct **604** has an uncompressed (undeformed), or pre-deployed, height and is configured to deform to one of a plurality of compressed (deformed), or deployed, heights. For example, the adjunct **604** can have an uncompressed height which is greater than the fired height of the staples **300** disposed within the staple cartridge **200** (e.g., the height (H) of the fired staple **300a** in FIG. 7A). That is, the adjunct **604** can have an undeformed state in which a maximum height of the adjunct **604** is greater than a maximum height of a fired staple (e.g., a staple that is in a formed configuration).

In use, once the surgical stapling and severing device, like device **100** in FIG. 1, is directed to the surgical site, tissue is positioned between the anvil **102** and the stapling assembly **600** such that the anvil **102** is positioned adjacent to a first side of the tissue and the stapling assembly **600** is positioned adjacent to a second side of the tissue (e.g., the tissue can be positioned against the tissue-contacting surface **604a** of the adjunct **604**). Once tissue is positioned between the anvil **102** and the stapling assembly **600**, the surgical stapler can be actuated, e.g., as discussed above, to thereby clamp the tissue between the anvil **102** and the stapling assembly **600** (e.g., between the tissue-compression surface **102a** of the anvil **102** and the tissue-contacting surface **604a** of the adjunct **604**) and to deploy staples from the cartridge through the adjunct and into the tissue to staple and attach the adjunct to the tissue.

As shown in FIG. 7A, when the staples **300** are fired, tissue (T) and a portion of the adjunct **604** are captured by the fired (formed) staples **300a**. The fired staples **300a** each define the entrapment area therein, as discussed above, for accommodating the captured adjunct **604** and tissue (T). The entrapment area defined by a fired staple **300a** is limited, at least in part, by a height (H) of the fired staple **300a**.

Referring to FIG. 7B, the adjunct **604** may have pores **632** with a median pore size of about 0.025 mm^3 to about 0.300 mm^3 , such as about 0.022 mm^3 . In some examples, the adjunct **604** may have one or more struts **634** between the pore **632** that provide support and strength to the adjunct **604**. In particular, the adjunct **604** may include a plurality of struts **634**, having a median strut thickness ST of about 0.025 mm to about 0.300 mm , such as about 0.08 mm .

FIG. 8 illustrates a perspective view of a staple cartridge assembly **600** with an adjunct **604** and a staple cartridge **200**. The adjunct **604** has a tissue contacting surface **604a**, a proximal end **604c**, and a distal end **604b**. The adjunct **604**

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may include a slot/slit **808** separating or partially separating two parallel portions of the adjunct **604**. In one example, adjunct **604** may include a slot **808** separating two parallel portions of the adjunct **604**, while in another example, adjunct **604** may include a slit **808** separating two parallel portions of the adjunct **604** and also one or more bridges (e.g., five bridges) **802** connecting the two parallel portions of the adjunct **604**. At least one bridge has a length in the longitudinal direction of about 0.035 inches to about 0.046 inches. The adjunct **604** has a length L of about 40 mm to about 80 mm, such as about 60 mm to about 65 mm, about 66.04 mm to about 66.3 mm, about 45 mm to about 55 mm, or about 51.12 mm to about 51.38 mm. The adjunct **604** has a width W of about 8 mm to about 12 mm, such as about 9.75 mm to about 10.25 mm or about 10.025 mm to about 10.035 mm. The adjunct **604** may also have a thickness or height TH of about 2.5 mm to about 3.5 mm, such as about 2.85 mm to about 3.15 mm or about 2.95 mm to about 3.05 mm.

The cartridge **200** has a height CH of about 6.3 mm to about 8.1 mm, a width CW of about 8.9 mm to about 14 mm, and a length CL of about 80 mm to about 90 mm such as about 86.7 mm.

The staple cartridge **200** may include one or more raised ledges **804** along one or more sides of the adjunct **604** to help align the adjunct **604** on the deck of the staple cartridge **200**. Although not shown in FIG. 8, the staple cartridge **200** may also include an adhesive or buttress adhesive material to attach the adjunct **604**. The adjunct **604** may be attached to the cartridge **200** with about 100 mg to about 120 mg of the adhesive or buttress adhesive material.

Adjusting Mechanical Properties Through Stimuli-Responsive Materials

The strength required to retain staples, sutures, screws, and the like, may conflict with the requirements for endoscopic deployment. As described above, the end effector **106** (shown in FIG. 1) comprising the cartridge **200** and adjunct **604** is closed or substantially closed for inserting through a trocar to the delivery site. As the implant strength increases (for maintaining staples and the hemostatic seal), there is an increased force required to compress the implant for insertion through the trocar. Therefore, the surgical adjunct **604** needs to have adjustable mechanical properties such that the material is sufficiently compressible during delivery through the trocar but increases strength at the delivery site to retain staples, sutures, and screws and maintain a hemostatic seal during the recovery period.

As described herein, the surgical adjunct **604** can be tuned for the particular purpose before, during, and after surgical procedures and have one or more of the functional groups described below that modulate the mechanical properties. In particular, the adjunct **604** described has one or more stimuli-responsive functional groups that respond to certain stimulus either ex vivo or in vivo, such that the compressibility of the adjunct **604** can be modulated during delivery and after firing.

FIG. 9A is a side view of the end effector **106** in a delivery configuration, where the adjunct **604** is compressible between the anvil **102** and the cartridge **200**. As illustrated, the end effector **106** can include a staple cartridge **200** and an adjunct **604** that is bioabsorbable can include a shape-memory polymer **624** that is compressible in a delivery configuration. In certain embodiments, the adjunct **604** is releasably retained on the cartridge **200**. As illustrated in FIG. 9A, when the end effector **106** closes around tissue T, the adjunct **604** can compress from a thickness of the uncompressed adjunct UT to a thickness of compressed adjunct CT in response to a variation in tissue thickness.

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FIG. 9B is a side view of the end effector **106** after firing of staples **300**, where the adjunct **604** is no longer compressed between the anvil **102** and the cartridge **200**. As shown, the adjunct **604** is released from the cartridge **200** after firing. The adjunct **604** is shown to maintain the thickness of the compressed adjunct CT and the thickness of uncompressed adjunct UT from pre-firing in FIG. 9A.

In some embodiments, the adjunct **604** can undergo a change in mechanical properties before firing of staples such that the uncompressed portion CT and compressed portion CT will be maintained after firing and releasing adjunct **604** and tissue T. In other embodiments, the adjunct **604** can undergo a change in mechanical properties after firing and within a predetermined time period. In such an example, the uncompressed portion CT and compressed portion CT can either expand or contract in response to the staple height and the tissue thickness as the tissue heals. As would be appreciated by those of skill in the art, a change in mechanical properties of the adjunct **604** can be advantageous to ensure a hemostatic seal during changes in tissue inflammatory responses and throughout the healing process.

To control mechanical properties of the bioabsorbable material, the adjunct **604** includes shape-memory polymer **624** that can swell via reversible transformations from linear polymer systems **625** to non-linear polymer systems **626**, as shown in FIGS. 9C and 9D. Stimuli-responsive functional groups can allow for transitions from linear polymer systems to complex architectures such as star, cyclic, or hyper-branched systems. The transition between linear polymers **625** to non-linear polymers **626** can include bonding among adjacent functional groups along the same polymer strand, as shown by the dashed lines in FIGS. 9C and 9D. Alternatively, or in addition thereto, the bonding can be cross-linking among functional groups along adjacent polymer strands, as shown by the dotted lines between the primary strand of FIG. 9C and a secondary strand in FIG. 9D (secondary polymer strand is not illustrated in FIG. 9C for clarity). The shape-memory polymer may include a backbone including one or more polymers selected from polyurethane, polyether urethane, polyester urethane, polyester urea, polyester, polycarbonate, polyorthoester, polyanhydride, polyesteramide, polyphosphazenes, polyphosphoesters, polysaccharides, and/or polyoxaester.

As would be understood by one of skill in the art, the polymer backbone may be further functionalized after formation of the backbone. For instance, a polyurethane backbone can be synthesized via a reaction between an isocyanate and a polyol. After formation of the polyurethane backbone, additional functional groups may be added. Alternatively, or in addition thereto, the polymer backbone precursors can be previously functionalized with one or more functional groups. As a non-limiting example of such a case, an isocyanate functionalized with one functional group may react with a polyol functionalized with a second functional group such that the polyurethane backbone is formed with pendant functional groups in the same step.

In certain embodiments, the shape-memory polymer **624** can be configured to adjust mechanical properties either through physical expansion (swelling) or a material phase change. The mechanical properties, such as the compression strength (compressing the polymer), tensile strength (stretching the polymer), flexural strength (bending of the polymer), torsional strength (twisting of the polymer), impact strength (under the effects of direct hammering or firing), tear resistance, ultimate elongation, and/or Young's modulus (ratio of stress to strain), can be configured to increase after releasing the shape-memory polymer **624**

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from the staple cartridge **200**. The functional groups can undergo reversible bonding with adjacent functional groups via stimulation comprising at least one of heat, light, water, electrical, magnetic, electromagnetic, ultrasound, pH, and the like. As would be appreciated by one of skill in the relevant art, each functional group can be selective to transition upon only one type of stimulation or may be able to transition with a combination of stimuli.

The mechanical properties can be reversed when using dampening implants, for instance, when using an adjunct **604** that needs to be sutured in vivo. To prevent damage during implant manipulation and suturing procedures, the shape-memory polymer **624** can be configured to have increased mechanical properties, but can be specifically tailored to retain a rubbery state (to prevent tearing) upon exposure to biological conditions (increased temperatures, pH, and the like).

In some embodiments, the shape-memory polymer **624** can be programmed via temperature, such that the adjunct **604** is deformed by a maximum change in compression (e_{max}) with a constant loading rate (e.g., 0.01 s^{-1}) at a programming temperature (T_D) and subsequently cooled to a loading temperature (T_L) while holding at the e_{max} . Outside of potential relaxation (Δe), the strain is recovered once raised to the recovery temperature (T_R). In general, the adjunct **604** may be designed such that the glass transition temperature (T_g) is between T_L and T_R . In addition, the adjunct **604** may also be designed such that the transition occurs through effective plasticization or degradation such that the effective T_g depresses and allows for recovery with a liquid emersion step prior to deployment. Liquid emersion may be water or any in vivo fluids within the body at the delivery site.

In certain other embodiments, the shape-memory polymer **624** can be designed around light mediation such that bonds of the functional groups are created or broken upon exposure to differing wavelengths of light. Activation or deactivation of the bonds within these systems can be done with a transmittable wavelength through the tissue or within the endoscopic procedure.

Temperature-triggered architectural transformations can be achieved in the present system through the use of thermally labile functional groups. Example thermally labile functional groups include Diels-Alder adducts and/or azo groups. For instance, Diels-Alder linkages based on a furan-maleimide reaction can be formed at relatively low temperatures ranging from about room temperature ($\sim 25^\circ \text{C}$.) to about 60°C . The furan-maleimide reaction can be reversed at higher temperatures, such as equal to or greater than approximately 90°C . The retro-cycloaddition of the furan-maleimide reaction generates free furan and maleimide moieties. In general, for the Diels-Alder thermal transition, the increase in mechanical properties of the adjunct **604** can occur in the range of about 34°C . to about 60°C . such as in the range of about 34°C . to about 40°C .

In some embodiments, to achieve a change in mechanical properties under a Diels-Alder reaction, the backbone of the shape-memory polymer **624** can include a diene and a dienophile moiety. The diene and a dienophile moiety can be incorporated in the polymer's monomer backbone or as pendant groups.

In some examples, the shape-memory polymer **624** can include a suitable diene, such as substituted or unsubstituted alkene. In some embodiments, the suitable diene can include, without limitation, furans, thiophenes, or pyrroles. Without intending to be bound, some example dienes include, without limitation, substituted or unsubstituted 1,2-

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propadiene, isoprene, 1,3-butadiene, 2,4-octanedione, 1,5-cyclooctadiene, norbornadiene, 2-pyrone, dicyclopentadiene, 1H-pyrrole-2-carboxylic acid, 1H-pyrrole-3-carboxylic acid, 3,5-dimethyl-1H-pyrrole-2-carboxylic acid, 1,5-dimethyl-1H-pyrrole-2-carboxylic acid, 2,4,5-trimethyl-1H-pyrrole-3-carboxylic acid, 5-phenyl-1H-pyrrole-2-carboxylic acid, 2,4-dimethyl-1H-pyrrole-3-carboxylic acid, 2,5-dimethyl-1H-pyrrole-3-carboxylic acid, 3-methyl-1H-pyrrole-2-carboxylic acid, 5-(3,4-dimethylphenyl)-2-methyl-1H-pyrrole-3-carboxylic acid, 1-methyl-1H-pyrrole-2-carboxylic acid, 2-methyl-1H-pyrrole-3-carboxylic acid, furan-2-carboxylic acid, furan-3-carboxylic acid, 2-(furan-2-yl) acetic acid, 3-(5-methylfuran-2-yl) propanoic acid, 5-ethylfuran-2-carboxylic acid, 5-isobutyl-2-methylfuran-3-carboxylic acid, 4,5-dimethylfuran-2-carboxylic acid, thiophene-2-carboxylic acid, 4,5-dimethylthiophene-2-carboxylic acid, 3-methylthiophene-2-carboxylic acid, 5-methylthiophene-2-carboxylic acid, 5-phenylthiophene-2-carboxylic acid, 2-(thiophen-2-yl) acetic acid, thiophene-3-carboxylic acid, 2-(thiophen-3-yl) acetic acid, 5-ethylthiophene-2-carboxylic acid, and 5-methyl-4-phenylthiophene-3-carboxylic acid.

In some embodiments, the diene undergoes the Diels-Alder cycloaddition reaction with a suitable dienophile that can include either a substituted or unsubstituted alkene or alkyne. In some embodiments, a suitable dienophile can include, without limitation, substituted or unsubstituted maleimide, acrolein, methyl vinyl ketone, acrylic acid, methyl acrylate, acrylamide, acrylonitrile, methyl acrylate, dimethyl maleate, dimethyl fumarate, maleic anhydride, Maleonitrile, butenolide, alpha-methylene gamma-butyrolactone, N-methylmaleimide, N-ethylmaleimide, dimethyl acetylene dicarboxylate, 6-maleimidohexanoic acid, 2-butenal, 2-Maleimidoacetic acid, 3-Maleimidopropionic acid, 3-Maleimidobenzoic acid, 3-(2,5-Dioxopyrrol-1-yl) hexanoic acid, 4-Maleimidobutyric acid, 4-Maleimidobenzoic acid, 4-(2,5-Dioxopyrrol-1-yl) hexanoic acid, 4-(2,5-dioxo-2,5-dihydro-pyrrol-1-yl)-benzoic acid, 5-Maleimidopentanoic acid, 6-Maleimidohexanoic acid, 6-(3-methyl-2,5-dioxopyrrol-1-yl) hexanoic acid, 6-(2,5-dioxopyrrol-1-yl)-2-methylhexanoic acid, 6-(2,5-Dioxopyrrol-1-yl)-4-methylhexanoic acid, 7-(2,5-Dioxo-2,5-dihydro-1H-pyrrol-1-yl) heptanoic acid, 9-(2,5-dioxopyrrol-1-yl) nonanoic acid, 10-(2,5-dioxopyrrol-1-yl) decanoic acid, 11-Maleimidoundecanoic acid, 13-(2,5-Dioxo-2,5-dihydro-1H-pyrrol-1-yl)tridecanoic acid, N-(Carboxyheptyl) maleimide, N-(4-Carboxy-3-hydroxyphenyl) maleimide, and α -Maleimidyl- ω -Carboxyl Poly(ethylene glycol).

In some embodiments, to achieve a change in mechanical properties under a thermal trigger, the backbone of the shape-memory polymer **624** can include an azo moiety. The azo moiety can be incorporated in the polymer's monomer backbone or as pendent groups. In general, for an azo thermal transition, the increase in mechanical properties of the adjunct **604** can occur in the range of about 60° C. to about 110° C. Alternatively, or in addition thereto, the azo moiety can be cleaved under light stimulation when the azo compound includes a substituted or unsubstituted aryl or heteroaryl groups within the azo molecule. Photoresponsive behaviors of the azo moiety within the shape-memory polymer **624** can be selected based on the azo compound. For example, an azo moiety with a trans-cis isomerization may be responsive to different wavelengths. For illustrative purposes, an azo moiety may experience photoresponsiveness in the trans-isomer at a wavelength ranging from about 350 nm to about 370 nm, whereas the cis-isomer may experience photoresponsiveness at a wavelength ranging

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from about 430 nm to about 460 nm. As would be appreciated by one of skill in the art, a shape-memory polymer containing one or more azo moieties may be tuned for a specific wavelength and/or temperature trigger.

In some examples, the shape-memory polymer **624** can include a suitable monazo, disazo, trisazo, polyazo, or azoic moiety. The monazo moieties can be schematically represented by the formula $Z-N=N-W$, where Z and W are substituted or unsubstituted aryl or heterocyclic groups. Diazo moieties contain two $-N=N-$ groups and can be symmetric or asymmetric. Polyazo moieties are characterized by the repetition of the azo group from three or more times in the same molecule. Suitable azo moieties can include, without limitation, substituted or unsubstituted triazolinones, poly(vinylcarbazole), 1,1'-Azobis(cyclohexanecarbonitrile) or ACHN, 4,4'-azobis(4-cyanovaleic acid), azobenzene, 4,4-dihydroxyazobenzene, p-azobenzenearsonate, 2,2'-azobis(2-amidinopropane) dihydrochloride, 4,4'-azobis(4-cyanopentanoic acid), azobisisobutyronitrile, azodicarbonamide, azoxy compounds, para-azoxyanisole, azoxybenzene, balsalazide, 3-Hydroxy-4-[(2-hydroxy-5-methylphenyl) azo]-1-naphthalenesulfonic acid ("calmagite"), diethyl azodicarboxylate, diimide, diisopropyl azodicarboxylate, 4,4'-dinitro-3,3'-diazonofuroxan, 1,3-diphenyltriazene, disodium 4,4'-dinitrostilbene-2,2'-disulfonate, fazadinium bromide, 4H-1,2,4-triazole-3,4,5-triazine with 5,5'-(1,2-diazenediyl)bis[2H-tetrazole] ("G2ZT"), glycoazodyes, methylazoxymethanol, methylazoxymethanol acetate, olsalazine (also known under trade name "Dipentum"), phenazopyridine, 3-phenylazoacetylacetone, 7,18-bis(4-phenyldiazenylphenyl)-7,18-diazaheptacyclo[14.6.2.22.5.03.12.04.9.013.23.020.24] hexacosal(23), 2,4,9,11,13,15,20 (24), 21,25-decaene-6,8,17,19-tetrone ("pigment red 178"), potassium azodicarboxylate, 4-[(E)-{4-formyl-5-hydroxy-6-methyl-3-[(phosphonoxy)methyl]pyridin-2-yl}diazenyl] benzene-1,3-disulfonic acid ("PPADS"), 6-methyl-2-(phenylazo)-3-pyridinol ("SIB-1757"), 4-phenyldiazenylphenol ("solvent yellow 7"), sulfasalazine, tetramethylazodicarboxamide, disodium 3-hydroxy-4-[(2-arsonophenyl)diazenyl] naphthalene-2,7-disulfonate ("thorin"), and 1,3,5-tri (p-glycosyloxyphenyl azo)-2,4,6-trihydroxybenzene ("Yariv reagent").

In some embodiments, to achieve a change in mechanical properties under photo-stimulation, the backbone of the shape-memory polymer **624** can include a styrylpyrene moiety. The styrylpyrene moiety can be incorporated in the polymer's monomer backbone or as pendent groups. Photoresponsive behaviors of the styrylpyrene moiety within the shape-memory polymer **624** can be selected to tune the wavelength of light required to initiate the change in mechanical properties. For instance, the shape-memory polymer **624** having a styrylpyrene moiety can be configured to undergo reversible transition of mechanical properties upon exposure to a wavelength of light ranging from about 310 nm to about 450 nm.

In some examples, the shape-memory polymer **624** can include a suitable styrylpyrene moiety. Suitable styrylpyrene moieties can include, without limitation, substituted or unsubstituted 1-styrylpyrene, phenanthracene, 3,4-benzopyrene, 1,2:5,6-dibenzanthracene, 1,2-benzanthracene, 7,12-dimethylbenzanthracene ("DMBA"), 1-[(1R,2S,4R)-5,6-dimethyl-2-bicyclo[2.2.1] heptanyl] pyrene, 1-(2-phenylethenyl) acenaphthylene, 2-(2-phenylethyl) dibenzofuran, 3-(2-phenylsulfanylethyl)-2-thia-3-azatricyclo[6.3.1.04,12] dodeca-1(11),4,6,8(12),9-pentaene, 1-(2-phenylethynyl) phenanthrene, 4-(2-phenylethenyl) pyrene, 2-(pyridin-2-ylmethoxy)-1,10-phenanthroline, 1-(3-

phenylpropyl) imidazo[2,1-b][1,3] benzothiazole, trimethyl-(4-pyren-1-ylphenyl) silane, 1-methyl-2-methylidene-6-(1-methylnaphthalen-2-yl)-3-prop-2-enyl-3,3a-dihydro-1H-acenaphthylene, and the like.

In some embodiments, to achieve a change in mechanical properties under photo-stimulation, the backbone of the shape-memory polymer **624** can include an ortho-nitrobenzyl moiety. The ortho-nitrobenzyl moiety can be incorporated in the polymer's monomer backbone or as pendent groups. In general, the shape-memory polymer **624** having an ortho-nitrobenzyl moiety is configured to reversibly transition upon exposure to a wavelength of light ranging from about 310 nm to about 440 nm.

In some examples, the shape-memory polymer **624** can include a suitable ortho-nitrobenzyl moiety. Suitable ortho-nitrobenzyl moieties can include, without limitation, substituted or unsubstituted [4-[(2-nitrophenyl)methyl]phenyl] methanediol, 1-(4,5-dimethoxy-2-nitro-phenyl)-but-3-ene-1-ol, 4-((1-(4,5-dimethoxy-2-nitrophenyl) but-3-en-1-yl)oxy)-4-oxobutanoic acid, 3-((2-acryloyloxymethyl-2-hydroxymethyl) propionyl) methyl-2-nitrobenzyl, 4-cyano-4-(phenylcarbonothioylthio) pentanoate ("ANCP"), 2-nitrobenzyl cyclohexylcarbamate, 2-[(2-nitrobenzyl)oxy]-1H-isindole-1,3 (2H)-dione, 2-[(2-Nitrophenyl)methylthio]-1,3-benzoxazole, 2-(2-Methyl-5-nitroimidazol-1-yl) ethyl thiophene-2-carboxylate, (E)-3-(2-chlorophenyl)-N-cyclopentyl-2-propenamide, (2E)-N-(5-chloropyridin-2-yl)-3-(2-methoxyphenyl) acrylamide, N-[(2-nitrophenyl) methylideneamino] thiophene-2-carboxamide, N-cyclohexyl-3-(2-methoxyphenyl) prop-2-enamide, (2E)-N-cyclo heptyl-3-(2-nitrophenyl) prop-2-enamide, 2-(2-nitrophenyl)-N-(thiophen-2-ylmethyl) acetamide, N-(3,5-dichloro-4-methylpyridin-2-yl)-2-(1-oxidopyridin-1-ium-2-yl) sulfanylacetamide, 1-(2-Nitrophenyl)-3-phenyl-2-thiourea, N-bicyclo[2.2.1] hept-2-yl-2-(2-nitrophenyl) acetamide, 2-[2-(2-Chloroanilino)-2-oxoethyl] sulfanylbenzoic acid, N-(2,5-dimethylphenyl)-3-methyl-1-oxo-3,4-dihydro-1H-isochromene-3-carboxamide, [(2-nitrophenyl) amino]-N-(1,3,4-thiadiazol-2-yl) carboxamide, 2-cyano-N-(2-methylcyclohexyl)-3-(5-methylthiophen-2-yl) prop-2-enamide, (E)-2-cyano-N-(2-methylcyclohexyl)-3-(3-methylthiophen-2-yl) prop-2-enamide, (E)-1-(2-nitrophenyl)-N-phenylmethoxymethanimine, N-(2-nitrophenyl)-N'-pyridin-2-ylurea, 2-(2,4-dimethyl-6-nitrophenoxy)-N-(5-methyl-1,2-oxazol-3-yl) propanamide, 2-(2-Nitrophenoxy)-1-(2-phenylpyrrolidin-1-yl) ethanone, 4-Chloro-1-[2-(4-methoxyphenoxy)ethylsulfanyl]-2-nitrobenzene, 2-[2-Oxo-2-(thiophen-2-ylmethylamino)ethyl] sulfanylbenzoic acid, 1-(2-nitrophenyl)-N-phenylmethoxymethanimine, N-(2-chloropyridin-3-yl)-2-[(2-nitrothien-3-yl)thio] acetamide, 2-(2-Nitrothiophen-3-yl) sulfanyl-1-pyrrolidin-1-ylethanone, (2-nitrophenyl)methyl N-[1-(2-hydroxyethoxy)-5-methylhexan-3-yl] carbamate, and the like.

In some embodiments, to achieve a change in mechanical properties, the backbone of the shape-memory polymer **624** can be a coumarin moiety. The coumarin moiety can be incorporated in the polymer's monomer backbone or as pendent groups. Coumarin moieties undergo a reversible [2 π s+2 π s] cycloaddition reaction upon irradiation with specific wavelengths in the UV region, which is applied to impart intrinsic healability, shape-memory, and reversible properties into polymers. During photoradiation, four different types of coumarin dimers are formed: anti head-to-head, anti head-to-tail, syn head-to-head, and syn head-to-tail.

In some examples, the shape-memory polymer **624** can include a suitable coumarin or coumarin-derivative moiety including dihydrofurano coumarins, furano coumarins, pyrano coumarins, phenyl coumarins, and bicoumarins. Suitable coumarin and/or coumarin-derivative moieties can include, without limitation, substituted or unsubstituted 2H-1-benzopyran-2-one ("coumarin"), 2-(Dimethylamino) ethyl methacrylate (DMAEMA) 7-(2-methacryloyloxyethoxy)-4-methylcoumarin (CMA), poly(DMAEMA-co-CMA). Dimethylaminoethyl acrylate (DMAEA), 6-Iodo-2H-chromen-2-one, 4-Hydroxycoumarin, 3-Hydroxycoumarin, 6-Methoxycoumarin, 4-Trimethylsiloxy-coumarin, and the like. In some embodiments, the shape-memory polymer **624** may be functionalized with coumarinyl end groups such that a single polymer strand can undergo a photodimerization.

In some embodiments, to achieve a change in mechanical properties, the backbone of the shape-memory polymer **624** can be an anthracene moiety. The anthracene moiety can be incorporated in the polymer's monomer backbone or as pendent groups. In general, anthracene groups undergo [4+4] photo-dimerization when irradiated by UV light ($\lambda > 300$ nm) and can be reversed to the original monomers via exposure to a higher energy UV light ($\lambda < 300$ nm).

In some examples, the shape-memory polymer **624** can include a suitable anthracene or anthracene-derivative moiety. Suitable anthracene and/or anthracene-derivative moieties can include, without limitation, substituted or unsubstituted Benzo[a]pyrene, Phenothiazine, Anthranol, Dibenzothiophene 5-oxide, 1,4,5-Trimethylnaphthalene, 4-Methyldibenzothiophene, Pyrene, 10-Methylacridin-9 (10H)-one, gamma-Fagarine, 9-Hydroxymethyl-10-methylanthracene, 2-Dodecylphenanthrene, Phenanthrene, 2-dodecyl-9,10-dihydro-, 2-Octyltriphenylene, 8a-Methyl-3,5-dimethylidene-3a,4,4a,6,7,8,9,9a-octahydrobenzo[f][1] benzo furan-2-one, Furanoeremophilane, 4-[(E)-2-(1-naphthyl) vinyl] biphenyl, 1,7-diazatricyclo[7.3.0.0.3,7] dodeca-3,5,9,11-tetraene-2,8-dione ("Pyrocoll"), (3R,4aR,8aR)-5,8a-dimethyl-3-prop-1-en-2-yl-2,3,4,4a, 7,8-hexahydro-1H-naphthalene ("alpha-Selinene"), 3,4-dihydro-2H-pyrimido [1,2-b][1,2] benzothiazole, and the like. In some embodiments, the shape-memory polymer **624** may be functionalized with anthracene end groups such that a single polymer strand can undergo a photodimerization.

In some embodiments, to achieve a change in mechanical properties, the backbone of the shape-memory polymer **624** can include a disulfide moiety. The disulfide moiety is a functional group with the formula R—S—S—R', where R and R' are either the same or different groups. The disulfide moiety can be incorporated in the polymer's monomer backbone or as pendent groups.

In general, disulfide moieties can be redox-responsive. Redox-active disulfide bonds are reversible and responsive to changes in the redox potential of the surrounding environment, and the formation or reduction of these disulfide bonds serve to increase the mechanical properties of the shape-memory polymer **624** after firing and delivery. The disulfide moieties can be intramolecular (oxidoreductases, allosteric disulfides, etc.) or mixed disulfides between a cysteine residue and a small-molecule thiol resulting in glutathionylated and cysteinylated adducts.

In some examples, the shape-memory polymer **624** can include a suitable disulfide moiety including, without limitation, substituted or unsubstituted diamines with disulfide groups, thiols with disulfide groups, or inimers with disulfide groups. Suitable disulfide moieties can include, without

limitation, substituted or unsubstituted thioredoxin disulfide, 2-(2'-bromoisobutyryloxy)ethyl-2"-methacryloyloxyethyl disulfide, and the like.

In some embodiments, to achieve a change in mechanical properties, the backbone of the shape-memory polymer **624** can be a diselenide moiety. The diselenide moiety can be incorporated in the polymer's monomer backbone or as pendent groups.

In general, diselenide moieties can be redox-responsive. Selenolate-diselenide equilibria are the same as their sulfur-containing counterparts, thiolate-diselenide equilibria, and involve the reversible formation of diselenides from selenolates through the two-electron oxidation of the selenolate group.

In some examples, the shape-memory polymer **624** can include a suitable diselenide moiety including, without limitation, substituted or unsubstituted selenols, diselenides, selenides, selenoxides, selenoketones, selenones, selenenic acids, or seleninic acids. Suitable diselenide moieties can include, without limitation, substituted or unsubstituted diselenocarbonate, selenocysteamine, selenocystine, selenocystine, glutathione, oxidized glutathione, and selenocystamine, and the like.

In some embodiments, to achieve a change in mechanical properties, the backbone of the shape-memory polymer **624** can be at least one of a diene and a dienophile moiety, a styrylpyrene moiety, an azo moiety, an ortho-nitrobenzyl moiety, a coumarin moiety, an anthracene moiety, a disulfide moiety, a diselenide moiety, or combinations thereof.

In general, these molecular changes via bond formation and breaking can be tailored to allow a macroscopic geometric change in the shape-memory polymer **624**. The transformation of the shape-memory polymer **624** can be thermal, photo, redox, or mechano-responsive.

In some embodiments, the functional groups along the shape-memory polymer **624** are more highly concentrated along a portion of the adjunct **604** so as to form a gradient of compression strength along a portion of the porous body **634**. In general, the adjunct **604** may have a compression strength of about 30 kPa to about 70 kPa, such as about 30 kPa to about 60 kPa (e.g., about 42 kPa), about 30 kPa to about 50 kPa, about 32.5 kPa to about 37.5 kPa. In some embodiments, the adjunct **604** may have a compression strength of about 15 to about 50 kPa in the second zone during delivery and a compression strength of about 30 kPa to about 70 kPa after delivery through the trocar, but before firing of staples. In order to test compression strengths, an adjunct **604** was placed in a humid warm environment at approximately 37° C., compressed to a first height, then a second height shorter than the first height, and then released back to the first height at which point the adjunct's compression strength was measured.

In any of the embodiments described herein, the adjunct **604** can be configured to reversible transition between an approximately linear polymer and an approximately non-linear polymer within approximately 0.01 seconds to approximately 15 minutes such that the swelling of the shape-memory polymer **624** occurs between the time of deployment through the trocar and firing of the staples, including various surgical procedure time delays. In some cases, the stimulation may cause the reversible transition to occur much faster, such that the shape-memory polymer **624** reaches substantial swelling and compression strength within seconds of being delivered through the trocar (e.g., within approximately 1 second, 2 seconds, 3 seconds, 4 seconds, 5 seconds, 10 seconds, 15 seconds, 20 seconds, 30 seconds, and any time in between). In other embodiments,

the shape-memory polymer **624** may be configured to reach substantial swelling and compression strength within a delayed time period from exposure to stimulation (e.g., after approximately 30 seconds, after 45 seconds, after 1 minute, after 2 minutes, and any time in between).

In some embodiments, the adjunct **604** may have a lower compressive strength during delivery and the compressive strength or other mechanical properties only increase after exposure to stimulus after deployment. Alternatively, the adjunct **604** can be compressed to a small thickness having high compressive strength but better margins for delivery through a trocar.

In some examples, the adjunct **604** may have a tensile strength of about 30 kPa to about 90 kPa such as about 45 kPa to about 85 kPa or about 55 kPa to about 75 kPa during delivery. After delivery, the adjunct **604** may increase the tensile strength from about 30 to about 45 kPa, or from about 45 kPa to about 65 kPa, or from about 55 kPa to about 75 kPa after exposure to stimulus. In some examples, the adjunct **604** will have tensile strength of about 110 kPa to about 150 kPa during delivery, that can increase upon exposure to a stimulus to a range of about 140 kPa to about 220 kPa.

FIGS. **10A** and **10B** show top and side view of an adjunct **604** after firing staples **300** and cutting tissue **T**. As shown, adjunct **604** may split in two post firing. Where the adjunct **604** contacts with the tissue **T**, bumps **604e**, **604f** may form that correspond to the texture and thickness variation in the tissue. This means that adjunct **604** can adapt to different heights and compressions depending on the application.

FIG. **11** is a flowchart of a method **1100** for forming a surgical adjunct **604**, including a bioabsorbable material that itself includes a shape-memory polymer. The techniques for adjusting mechanical properties described herein may offer an added benefit of increasing adjunct strength and durability in combination with positive in vivo interactions (e.g., biocompatibility, wound healing, tissue integration, chemotherapy, anti-inflammatory, bone growth and integration, ligament and tendon repair, etc.) when the adjunct is delivered to the tissue site, as described herein. As such, the techniques described herein may allow the bioabsorbable material itself to aid in the healing process of the surrounding tissue. In addition, the embedding techniques described herein may offer an added benefit of preventing fibrous encapsulation of the foam cushion, and/or providing tunable release profiles for a variety of medical additives delivered to the tissue site.

Specifically with respect to FIG. **11**, method **1100** used for forming a bioabsorbable material (e.g., a foam) having shape-memory polymer may include adding, to a polyurethane polymer, a functional group that can reversibly cross-link upon exposure to stimulation (step **1102**). The functional group can be at least one of a diene and a dienophile moiety, a styrylpyrene moiety, an azo moiety, an ortho-nitrobenzyl moiety, a coumarin moiety, an anthracene moiety, a disulfide moiety, a diselenide moiety, or combinations thereof. Method **1100** can next include chemically bonding the polyurethane polymer and functional groups to form the shape-memory polymer (step **1104**). In some examples, the bioabsorbable material may include a polyurethane, or alternatively, may include polyether urethane, polyester urethane, polyester urea, polyester, polycarbonate, polyorthoe-ster, polyanhydride, polyesteramide, polyphosphazenes, polyphosphoesters, polysaccharides, and/or polyoxaester. Method **1100** also includes exposing the shape-memory polymer to a stimulation comprising at least one of heat, light, water, electrical, magnetic, electromagnetic, ultra-

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sound, and pH (step 1106). After exposure to the stimulation, method 1100 includes increasing a height of the shape-memory polymer to greater than a delivery height (step 1108). Method 1100 can end after step 1108 or can optionally include adding a medical additive to the porous body. In such examples, the medical additives may include medicants to treat pain and/or promote wound healing, tissue growth, infection reduction, and the like.

As will be appreciated by one skilled in the art, The embodiments described above are cited by way of example, and the present invention is not limited by what has been particularly shown and described hereinabove. Rather, the scope of the invention includes both combinations and sub combinations of the various features described hereinabove, as well as variations and modifications thereof which would occur to persons skilled in the art upon reading the foregoing description and which are not disclosed in the prior art.

In some examples, disclosed devices (e.g., end effector, surgical adjunct, and/or staple cartridges) and methods involving one or more disclosed devices may involve one or more of the following clauses:

Clause 1: A bioabsorbable material configured to be delivered to tissue, the material comprising: a shape-memory polymer compressible in a delivery configuration and configured to swell within a predetermined period of time; wherein the shape-memory polymer comprises one or more functional groups for reversible bonding between adjacent functional groups to transition between an approximately linear polymer and an approximately non-linear polymer upon exposure to a stimulation.

Clause 2: The bioabsorbable material of clause 1, wherein the shape-memory polymer transitions from the delivery configuration to a swollen configuration when exposed to a temperature in a range of about 34° C. to about 40° C.

Clause 3: The bioabsorbable material of clause 1, wherein the stimulation comprises at least one of heat, light, water, electrical, magnetic, electromagnetic, ultrasound, and pH.

Clause 4: The bioabsorbable material of clause 1, wherein the shape-memory polymer comprises a reaction product of a polyol and an isocyanate.

Clause 5: The bioabsorbable material of clause 1, wherein the functional groups comprise a diene and a dienophile moieties, and wherein the shape-memory polymer is configured to reversibly transition upon exposure to a temperature in a range of about 34° C. to about 40° C.

Clause 6: The bioabsorbable material of clause 1, wherein the functional groups comprise a styrylpyrene moiety, and wherein the shape-memory polymer is configured to reversibly transition upon exposure to a wavelength of light ranging from about 310 nm to about 450 nm.

Clause 7: The bioabsorbable material of clause 1, wherein the functional groups comprise an azo moiety, and wherein the shape-memory polymer is configured to reversibly transition upon exposure to a temperature above about 60° C.

Clause 8: The bioabsorbable material of clause 1, wherein the functional groups comprise an ortho-nitrobenzyl moiety, and wherein the shape-memory polymer is configured to reversibly transition upon exposure to a wavelength of light ranging from about 310 nm to about 440 nm.

Clause 9: The bioabsorbable material of clause 1, wherein the functional groups comprise a coumarin moiety, wherein the shape-memory polymer is configured to

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reversibly transition to a first compression strength upon exposure to a wavelength of light ranging from about 200 nm to about 260 nm; and wherein the shape-memory polymer is configured to reversibly transition to a second compression strength upon exposure to a wavelength of light ranging from about 350 nm to about 560 nm.

Clause 10: The bioabsorbable material of clause 1, wherein the functional groups comprise an anthracene moiety, and wherein the shape-memory polymer is configured to reversibly transition to the approximately non-linear configuration upon exposure to a wavelength of light greater than about 300 nm, and transition to the approximately linear configuration upon exposure to a wavelength of light less than about 300 nm.

Clause 11: The bioabsorbable material of clause 5, wherein the functional groups comprise at least one of a disulfide moiety and a diselenide moiety, and wherein the shape-memory polymer is configured to reversibly transition upon exposure to at least one of a change in temperature, a change in pH, a reactive oxygen species, or combinations thereof.

Clause 12: The bioabsorbable material of clause 1, wherein the predetermined period of time ranges from approximately 0.01 seconds to approximately 120 seconds.

Clause 13: The bioabsorbable material of clause 2, wherein the bioabsorbable material has a compression strength of about 50 kPa to about 90 kPa in the delivery configuration, and a compression strength of about 30 kPa to about 70 kPa in the swollen configuration.

Clause 14: The bioabsorbable material of clause 1, wherein the bioabsorbable material degrades according to a degradation profile in response to exposure to a fluid comprising at least one of a predetermined temperature, an enzyme-catalyst, and a predetermined pH.

Clause 15: The bioabsorbable material of clause 1, further comprising one or more medical additives configured to remain chemically bonded to the shape-memory polymer.

Clause 16: The bioabsorbable material of clause 15, wherein the one or more medical additives are further configured to be released to or approximate the tissue.

Clause 17: A bioabsorbable material configured to be delivered to tissue, the material comprising: a shape-memory polymer compressible in a delivery configuration and configured to swell within a predetermined period of time upon exposure to a stimulation, the shape-memory polymer comprising a polyurethane backbone and one or more functional groups for reversible bonding between adjacent functional groups to transition between an approximately linear polymer to an approximately non-linear polymer.

Clause 18: The bioabsorbable material of clause 17, wherein the stimulation comprises at least one of heat, light, water, electrical, magnetic, electromagnetic, ultrasound, and pH.

Clause 19: The bioabsorbable material of clause 1, wherein the predetermined period of time ranges from approximately 0.01 seconds to approximately 120 seconds.

Clause 20: The bioabsorbable material of clause 17, wherein the one or more functional groups comprise at least one of a diene and a dienophile moiety, a styrylpyrene moiety, an azo moiety, an ortho-nitroben-

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zyl moiety, a coumarin moiety, an anthracene moiety, a disulfide moiety, a diselenide moiety, or combinations thereof.

- Clause 21: A method to form a bioabsorbable material configured to be placed inside a body of a human, the method comprising the steps of: adding, to a polyurethane polymer, a functional group comprising at least one of a diene and a dienophile moiety, a styrylpyrene moiety, an azo moiety, an ortho-nitrobenzyl moiety, a coumarin moiety, an anthracene moiety, a disulfide moiety, a diselenide moiety, or combinations thereof; and chemically bonding the polyurethane polymer and functional groups to form a shape-memory polymer.
- Clause 22: The method of clause 21, further comprising the step of: exposing the shape-memory polymer to a stimulation comprising at least one of heat, light, water, electrical, magnetic, electromagnetic, ultrasound, and pH.
- Clause 23: The method of clause 22, wherein the shape-memory polymer is configured to reversibly transition between an approximately linear polymer to an approximately non-linear polymer.
- Clause 24: The method of clause 22, wherein the shape-memory polymer is compressible in a delivery configuration.
- Clause 25: The method of clause 21, wherein the bioabsorbable material comprises a delivery height ranging from about 0.01 mm to about 1 mm when in the delivery configuration.
- Clause 26: The method of clause 25, further comprising the step of: increasing a height of the shape-memory polymer to greater than the delivery height.
- Clause 27: The method of clause 21, further comprising the step of: exposing the bioabsorbable material to a fluid comprising at least one of a predetermined temperature, an enzyme-catalyst, and a predetermined pH such that the bioabsorbable material degrades according to a degradation profile.
- Clause 28: A bioabsorbable material configured to be delivered to tissue, the material comprising: a shape-memory polymer compressible in a delivery configuration and configured to swell within a predetermined period of time; wherein the shape-memory polymer comprises one or more functional groups for reversible bonding between adjacent functional groups to transition between an approximately linear polymer and an approximately non-linear polymer upon exposure to a stimulation.
- Clause 29: The material of clause 28, wherein the shape-memory polymer transitions from the delivery configuration to a swollen configuration when exposed to a temperature in a range of about 34° C. to about 40° C.
- Clause 30: The material of clauses 28 or 29, wherein the stimulation comprises at least one of heat, light, water, electrical, magnetic, electromagnetic, ultrasound, and pH.
- Clause 31: The material any of clauses 28-30, wherein the shape-memory polymer comprises a reaction product of a polyol and an isocyanate.
- Clause 32: The material of any of clauses 28-31, wherein the functional groups comprise a diene moiety and a dienophile moiety, and wherein the shape-memory polymer is configured to reversibly transition upon exposure to a temperature above in a range of about 34° C. to about 40° C.
- Clause 33: The material of any of clauses 28-31, wherein the functional groups comprise a styrylpyrene moiety,

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and wherein the shape-memory polymer is configured to reversibly transition upon exposure to a wavelength of light ranging from about 310 nm to about 450 nm.

- Clause 34: The material of any of clauses 28-31, wherein the functional groups comprise an azo moiety, and wherein the shape-memory polymer is configured to reversibly transition upon exposure to a temperature above about 60° C.
- Clause 35: The material of any of clauses 28-31, wherein the functional groups comprise an ortho-nitrobenzyl moiety, and wherein the shape-memory polymer is configured to reversibly transition upon exposure to a wavelength of light ranging from about 310 nm to about 440 nm.
- Clause 36: The material of any of clauses 28-31, wherein the functional groups comprise a coumarin moiety, wherein the shape-memory polymer is configured to reversibly transition to a first compression strength upon exposure to a wavelength of light ranging from about 200 nm to about 260 nm; and wherein the shape-memory polymer is configured to reversibly transition to a second compression strength upon exposure to a wavelength of light ranging from about 350 nm to about 560 nm.
- Clause 37: The material of any of clauses 28-31, wherein the functional groups comprise an anthracene moiety, and wherein the shape-memory polymer is configured to reversibly transition to the approximately non-linear configuration upon exposure to a wavelength of light greater than about 300 nm, and transition to the approximately linear configuration upon exposure to a wavelength of light less than about 300 nm.
- Clause 38: The material any of any of clauses 28-31, wherein the functional groups comprise at least one of a disulfide moiety and a diselenide moiety, and wherein the shape-memory polymer is configured to reversibly transition upon exposure to at least one of a change in temperature, a change in pH, a reactive oxygen species, or combinations thereof.
- Clause 39: The material of any of clauses 28-38, wherein the predetermined period of time ranges from approximately 0.028 seconds to approximately 120 seconds.
- Clause 40: The material of clauses 28 or 29, wherein the bioabsorbable material has a compression strength of about 50 kPa to about 90 kPa in the delivery configuration, and a compression strength of about 30 kPa to about 70 kPa in the swollen configuration.
- Clause 41: The material any of clauses 28-40, wherein the bioabsorbable material degrades according to a degradation profile in response to exposure to a fluid comprising at least one of a predetermined temperature, an enzyme-catalyst, and a predetermined pH.
- Clause 42: The material of any of clauses 28-41, further comprising one or more medical additives configured to remain chemically bonded to the shape-memory polymer, wherein the one or more medical additives are further configured to be released to or approximate the tissue.

What is claimed is:

1. A bioabsorbable material configured to be delivered to tissue, the material comprising: a shape-memory polymer compressible in a delivery configuration and configured to swell for a predetermined period of time comprising a range from approximately 0.01 seconds to approximately 120 seconds; wherein the shape-memory polymer comprises one or more functional groups for reversible bonding between adjacent functional groups to transition between an approxi-

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mately linear polymer and an approximately non-linear polymer upon exposure to a stimulation.

2. The bioabsorbable material of claim 1, wherein the shape-memory polymer transitions from the delivery configuration to a swollen configuration when exposed to a temperature in a range of about 34° C. to about 40° C.

3. The bioabsorbable material of claim 1, wherein the stimulation comprises at least one of heat, light, water, electrical, magnetic, electromagnetic, ultrasound, and pH.

4. The bioabsorbable material of claim 3, wherein the bioabsorbable material has a compression strength of about 50 kPa to about 90 kPa in the delivery configuration, and a compression strength of about 30 kPa to about 70 kPa in the swollen configuration.

5. The bioabsorbable material of claim 1, wherein the shape-memory polymer comprises a reaction product of a polyol and an isocyanate.

6. The bioabsorbable material of claim 1, wherein the functional groups comprise a diene and a dienophile moieties, and wherein the shape-memory polymer is configured to reversibly transition upon exposure to a temperature in a range of about 34° C. to about 40° C.

7. The bioabsorbable material of claim 6, wherein the functional groups comprise at least one of a disulfide moiety and a diselenide moiety, and wherein the shape-memory polymer is configured to reversibly transition upon exposure to at least one of a change in temperature, a change in pH, a reactive oxygen species, or combinations thereof.

8. The bioabsorbable material of claim 1, wherein the functional groups comprise a styrylpyrene moiety, and wherein the shape-memory polymer is configured to reversibly transition upon exposure to a wavelength of light ranging from about 310 nm to about 450 nm.

9. The bioabsorbable material of claim 1, wherein the functional groups comprise an azo moiety, and wherein the shape-memory polymer is configured to reversibly transition upon exposure to a temperature above about 60° C.

10. The bioabsorbable material of claim 1, wherein the functional groups comprise an ortho-nitrobenzyl moiety, and wherein the shape-memory polymer is configured to reversibly transition upon exposure to a wavelength of light ranging from about 310 nm to about 440 nm.

11. The bioabsorbable material of claim 1, wherein the functional groups comprise a coumarin moiety, wherein the shape-memory polymer is configured to reversibly transition to a first compression strength upon exposure to a wavelength of light ranging from about 200 nm to about 260 nm; and wherein the shape-memory polymer is configured to reversibly transition to a second compression strength upon exposure to a wavelength of light ranging from about 350 nm to about 560 nm.

12. The bioabsorbable material of claim 1, wherein the functional groups comprise an anthracene moiety, and wherein the shape-memory polymer is configured to reversibly transition to the approximately non-linear configuration upon exposure to a wavelength of light greater than about 300 nm, and transition to the approximately linear configuration upon exposure to a wavelength of light less than about 300 nm.

13. The bioabsorbable material of claim 1, wherein the bioabsorbable material degrades according to a degradation

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profile in response to exposure to a fluid comprising at least one of a predetermined temperature, an enzyme-catalyst, and a predetermined pH.

14. The bioabsorbable material of claim 1, further comprising one or more medical additives configured to remain chemically bonded to the shape-memory polymer.

15. The bioabsorbable material of claim 14, wherein the one or more medical additives are further configured to be released to or approximate the tissue.

16. A bioabsorbable material configured to be delivered to tissue, the material comprising: a shape-memory polymer compressible in a delivery configuration and configured to swell for a predetermined period of time upon exposure to a stimulation, wherein the predetermined period of time comprises a range from approximately 0.01 seconds to approximately 120 seconds, the shape-memory polymer comprising a polyurethane backbone and one or more functional groups for reversible bonding between adjacent functional groups to transition between an approximately linear polymer to an approximately non-linear polymer.

17. The bioabsorbable material of claim 16, wherein the stimulation comprises at least one of heat, light, water, electrical, magnetic, electromagnetic, ultrasound, and pH.

18. The bioabsorbable material of claim 16, wherein the one or more functional groups comprise at least one of a diene and a dienophile moiety, a styrylpyrene moiety, an azo moiety, an ortho-nitrobenzyl moiety, a coumarin moiety, an anthracene moiety, a disulfide moiety, a diselenide moiety, or combinations thereof.

19. A method to form a bioabsorbable material configured to be placed inside a body of a human, the method comprising the steps of: adding, to a polyurethane polymer, a functional group comprising at least one of a diene and a dienophile moiety, a styrylpyrene moiety, an azo moiety, an ortho-nitrobenzyl moiety, a coumarin moiety, an anthracene moiety, a disulfide moiety, a diselenide moiety, or combinations thereof; and chemically bonding the polyurethane polymer and functional groups to form a shape-memory polymer.

20. The method of claim 19, further comprising the step of: exposing the shape-memory polymer to a stimulation comprising at least one of heat, light, water, electrical, magnetic, electromagnetic, ultrasound, and pH.

21. The method of claim 20, wherein the shape-memory polymer is configured to reversibly transition between an approximately linear polymer to an approximately non-linear polymer.

22. The method of claim 20, wherein the shape-memory polymer is compressible in a delivery configuration.

23. The method of claim 19, wherein the bioabsorbable material comprises a delivery height ranging from about 0.01 mm to about 1 mm when in the delivery configuration.

24. The method of claim 23, further comprising the step of: increasing a height of the shape-memory polymer to greater than the delivery height.

25. The method of claim 19, further comprising the step of: exposing the bioabsorbable material to a fluid comprising at least one of a predetermined temperature, an enzyme-catalyst, and a predetermined pH such that the bioabsorbable material degrades according to a degradation profile.

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